DESIGN BASIS REPORT

ON PROPOSED STEELWORK FOR DOOR SUPPORT

CLIENT: SERUM INSTITUTE OF INDIA PVT. LTD

Contents

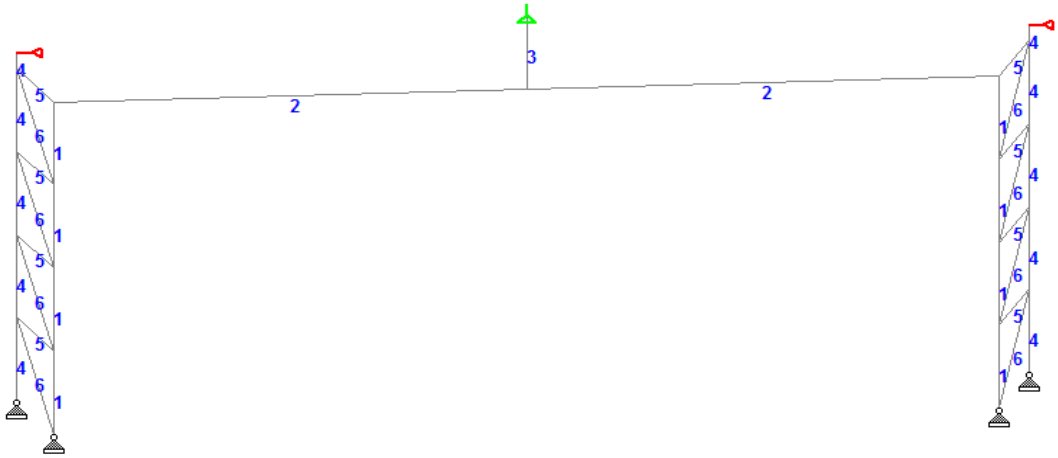
1. Introduction 3
2. Section properties 3
3. Load Data: 4
3.1 Door closed condition 4
3.2 Door open condition 5
4. Primary Load & Load Combination 5
5. Design & Analysis 6
5.1 Design & Analysis of steel frame 6
5.2 steel connections 45

1. Introduction

It is proposed to install a Mega-door of size 20mx 8.9m in an existing hanger PEB Shed. A structural steelwork frame is proposed for supporting Mega-door. Proposed columns are braced with existing PEB column to transfer wind load forces. The header beam is supported at mid-span by a hanger from existing structure so that the span of 20m is reduced to 10m.

2. Section properties

Prop	Section	Area (cm^2)	I_{yy} (cm^4)	I_{zz} (cm^4)	J (cm^4)	Material
1	ISMB250	47.600	335.000	5.13E 3	25.000	STEEL
2	ISMB300 D	117.400	6.72E 3	18E 3	50.696	STEEL
3	TUB1003.2	8.930	75.500	75.500	118.447	STEEL
4	Taper	58.400	800.717	42.5E 3	5.747	STEEL
5	ISMC75 FR	18.200	145.885	156.000	2.748	STEEL
6	ISMC75 FR	18.200	145.885	156.000	2.748	STEEL



Frame showing sections used

3. Load Data

The Loads are applied as per the input by Mega-door vendor. These loads are applied for two conditions as follows:

1) Door closed condition
2) Door open condition

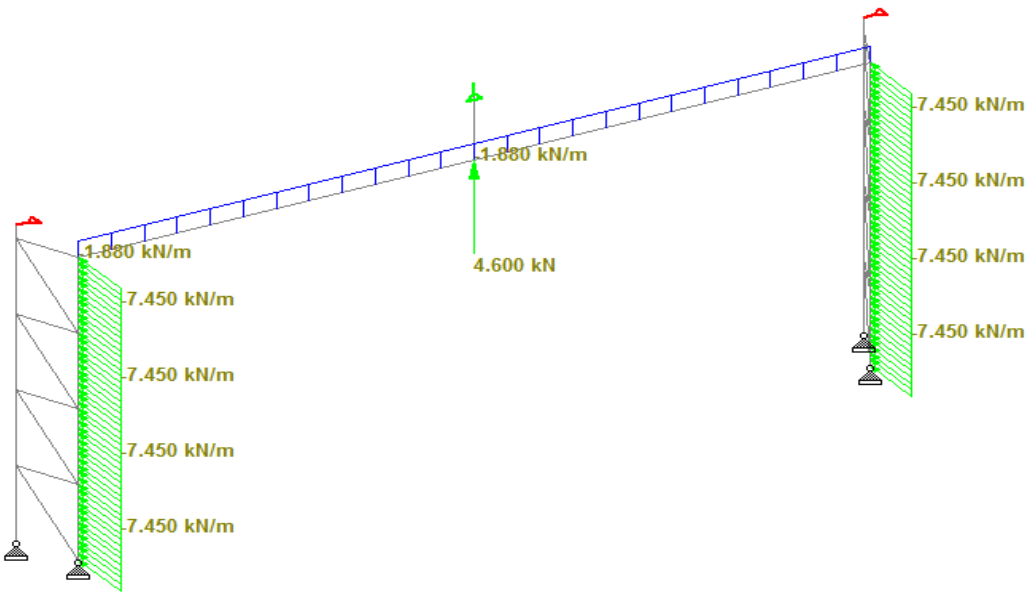
Axial load at gear motor 3.1 kN is applied directly on hanger.

3.1 Door closed condition

Vertical load: 4.6 kN upward applied on header box attachment (due to wind load and door weight)

Distributed vertical load: 37.5/20 = 1.88kN/m

Lateral load on columns: 13.4*5/9 = 7.45 kN/m over a length of column

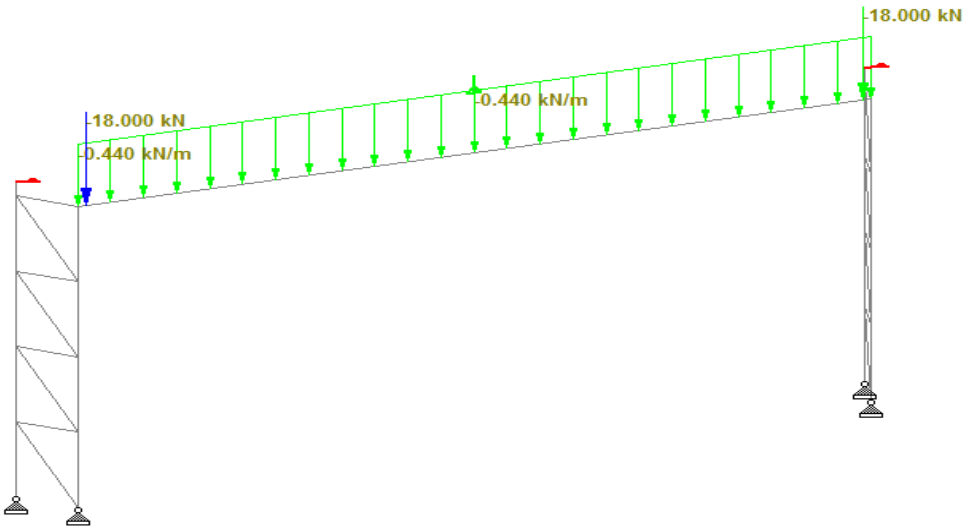


Loading on frame at Door Closed condition

3.2 Door open condition

Distributed vertical load: 8.7/20 = 0.44kN/m

Vertical point loads: 18 kN each at both ends of door



Loading on frame at Door Open condition

Self-Weight of Structural steel members is accounted in STAAD model.

All these loads are assumed to be equally distributed at all nodes of a frame. These loads are assigned in STAAD model as nodal point loads and distributed loads.

4. Primary Loads & Load Combinations:

PRIMARY LOADS

1. DL: Self weight of Structure
2. DC: Loading at door closed condition
3. DO: Loading at door open condition

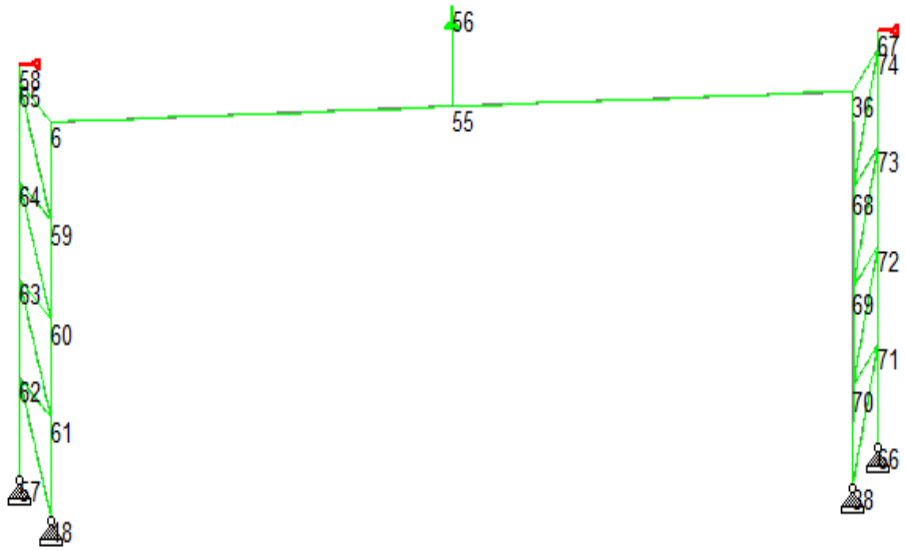
LOAD COMBINATIONS:

4. DL+DC
5. DL+DO

5. Design and Analysis:

5.1 Design & analysis of Steel Frame

The 3D modeling of the steel frame is done on STAAD. The dead load is applied in vertical direction and wind load is applied in lateral direction. The sections are designed for load combinations as mentioned above. The existing portal frame shall be checked for additional wind load on Megadoor.



Frame with node numbers

The support reactions are tabulated as below:

Node	L/C	Force-X kN	Force-Y kN	Force-Z kN	Moment-X kNm	Moment-Y kNm	Moment-Z kNm
18	1	0.163	8.151	-0.019	0.000	0.000	0.000
	2	19.014	-189.544	67.067	0.000	0.000	0.000
	3	0.147	19.305	-0.016	0.000	0.000	0.000
	4	19.176	-181.393	67.048	0.000	0.000	0.000
	5	0.310	27.456	-0.035	0.000	0.000	0.000
38	1	-0.163	8.151	-0.019	0.000	0.000	0.000
	2	-19.014	-189.544	67.067	0.000	0.000	0.000

	3	-0.147	19.305	-0.016	0.000	0.000	0.000
	4	-19.176	-181.393	67.048	0.000	0.000	0.000
	5	-0.310	27.456	-0.035	0.000	0.000	0.000
56	1	0.000	11.129	0.000	0.000	0.000	0.000
	2	0.000	9.788	0.000	0.000	0.000	0.000
	3	0.000	6.191	0.000	0.000	0.000	0.000
	4	0.000	20.918	0.000	0.000	0.000	0.000
	5	0.000	17.320	0.000	0.000	0.000	0.000
57	1	0.008	5.461	0.019	0.000	0.000	0.000
	2	-0.015	201.150	-0.017	0.000	0.000	0.000
	3	0.006	-0.000	0.016	0.000	0.000	0.000
	4	-0.007	206.611	0.002	0.000	0.000	0.000
	5	0.014	5.461	0.035	0.000	0.000	0.000
66	1	-0.008	5.461	0.019	0.000	0.000	0.000
	2	0.015	201.150	-0.017	0.000	0.000	0.000
	3	-0.006	0.000	0.016	0.000	0.000	0.000
	4	0.007	206.611	0.002	0.000	0.000	0.000
	5	-0.014	5.461	0.035	0.000	0.000	0.000

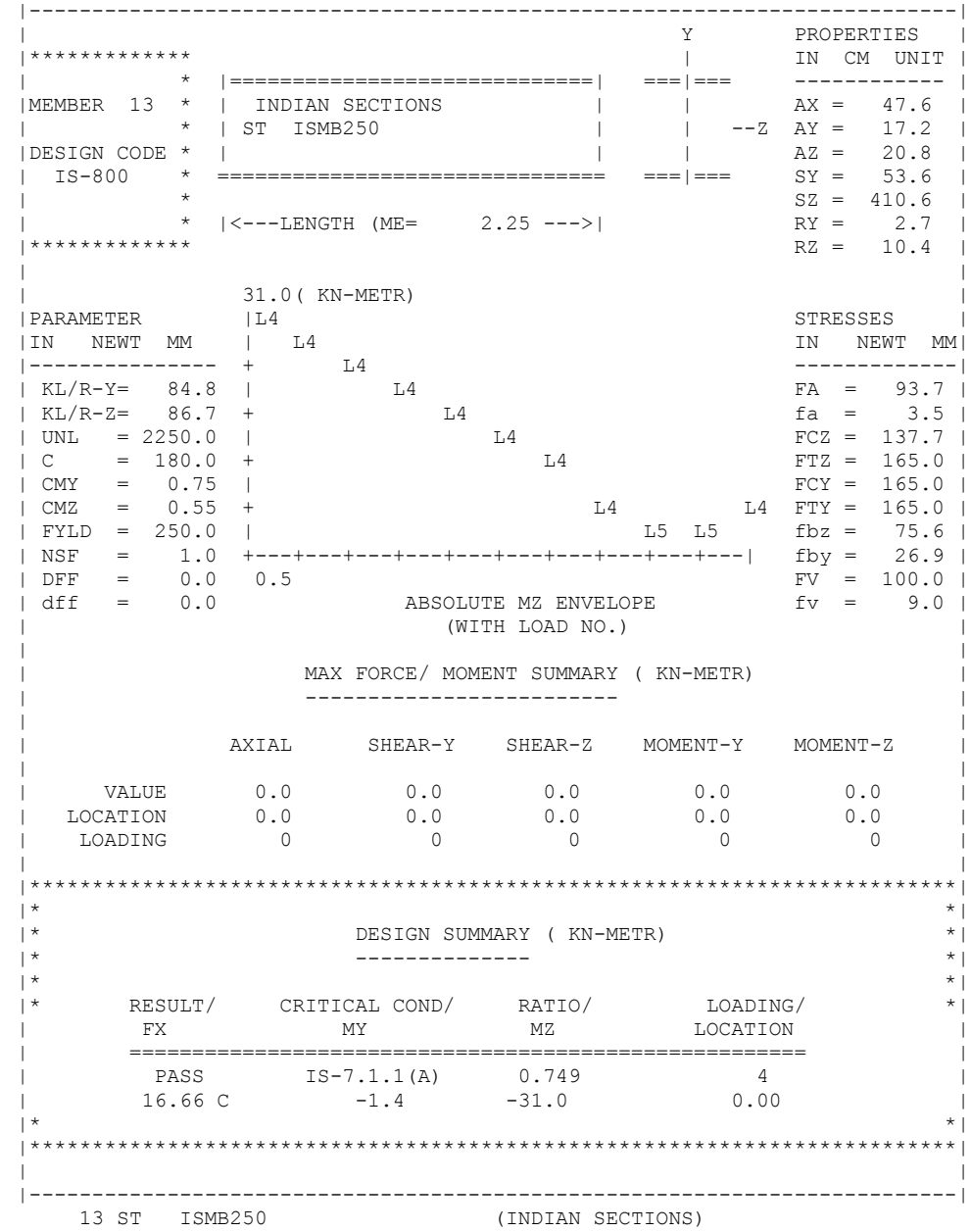
Nodal displacements of framing due to self- weight and other vertical and lateral load are tabulated below

Node	L/C	X-Trans mm	Y-Trans mm	Z-Trans mm	Absolute mm	X-Rotan rad	Y-Rotan rad	Z-Rotan rad
6	1	0.001	-0.054	0.040	0.068	-0.000	0.000	-0.000
	2	0.079	0.284	-9.022	9.026	0.002	-0.000	0.001
	3	0.001	-0.178	0.401	0.439	-0.000	0.000	-0.000
18	1	0.000	0.000	0.000	0.000	0.000	-0.000	0.000

	2	0.000	0.000	0.000	0.000	-0.006	0.006	-0.005
	3	0.000	0.000	0.000	0.000	0.000	-0.000	0.000
36	1	-0.001	-0.054	0.040	0.068	-0.000	-0.000	0.000
	2	-0.079	0.284	-9.022	9.026	0.002	0.000	-0.001
	3	-0.001	-0.178	0.401	0.439	-0.000	-0.000	0.000
38	1	0.000	0.000	0.000	0.000	0.000	0.000	-0.000
	2	0.000	0.000	0.000	0.000	-0.006	-0.006	0.005
	3	0.000	0.000	0.000	0.000	0.000	0.000	-0.000
56	1	-0.000	0.000	-0.365	0.365	-0.000	-0.000	0.000
	2	-0.000	0.000	-3.807	3.807	0.002	0.000	0.000
	3	-0.000	0.000	0.265	0.265	-0.000	-0.000	0.000
57	1	0.000	0.000	0.000	0.000	-0.000	-0.000	0.000
	2	0.000	0.000	0.000	0.000	-0.001	0.005	-0.000
	3	0.000	0.000	0.000	0.000	-0.000	-0.000	0.000
58	1	0.000	-0.022	0.052	0.057	0.000	-0.000	-0.000
	2	0.000	-0.696	-9.307	9.333	-0.001	0.001	0.000
	3	0.000	-0.000	0.438	0.438	0.000	-0.000	-0.000
66	1	0.000	0.000	0.000	0.000	-0.000	0.000	-0.000
	2	0.000	0.000	0.000	0.000	-0.001	-0.005	0.000
	3	0.000	0.000	0.000	0.000	-0.000	0.000	-0.000
67	1	0.000	-0.022	0.052	0.057	0.000	0.000	0.000
	2	0.000	-0.696	-9.307	9.333	-0.001	-0.001	-0.000
	3	0.000	-0.000	0.438	0.438	0.000	0.000	0.000

Following is the detailed steel framing design output for a frame based on IS 800

Steel Design (Track 2) Beam 13 Check 1



Steel Design (Track 2) Beam 49 Check 1

```
|-------------------------------------------------------------------------|
|                                                  Y        PROPERTIES    |
|*************                                     |        IN  CM  UNIT  |
|            *  |==============================|   ===|===     ------------ |
|MEMBER  49  *  |  INDIAN SECTIONS             |      |        AX =   47.6  |
|            *  | ST  ISMB250                  |      |   --Z  AY =   17.2  |
|DESIGN CODE *  |                              |      |        AZ =   20.8  |
|  IS-800    *  ================================   ===|===     SY =   53.6  |
|            *                                                  SZ =  410.6  |
|            *  |<---LENGTH (ME=     2.25 --->|                RY =    2.7  |
|*************                                                 RZ =   10.4  |
|                                                                         |
|                 31.0( KN-METR)                                          |
|PARAMETER        |L4                                           STRESSES    |
|IN   NEWT  MM    |   L4                                        IN   NEWT  MM|
|---------------  +       L4                                    -------------|
| KL/R-Y=   84.8  |           L4                                FA  =   93.7 |
| KL/R-Z=   86.7  +               L4                          fa  =    3.5 |
| UNL   = 2250.0  |                   L4                        FCZ =  137.7 |
| C     =  180.0  +                       L4                    FTZ =  165.0 |
| CMY   =   0.75  |                                             FCY =  165.0 |
| CMZ   =   0.55  +                           L4          L4      FTY =  165.0 |
| FYLD  =  250.0  |                                 L5  L5      fbz =   75.6 |
| NSF   =    1.0  +---+---+---+---+---+---+---+---+---+---|      fby =   26.9 |
| DFF   =    0.0   0.5                                        FV  =  100.0 |
| dff   =    0.0              ABSOLUTE MZ ENVELOPE            fv  =    9.0 |
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY ( KN-METR)                |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE       0.0          0.0        0.0         0.0         0.0    |
|   LOCATION       0.0          0.0        0.0         0.0         0.0    |
|    LOADING         0            0          0           0           0    |
|                                                                         |
|*************************************************************************|
|*                                                                        *|
|*                         DESIGN SUMMARY ( KN-METR)                      *|
|*                         --------------                                 *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY              MZ           LOCATION            |
|         ========================================================        |
|          PASS        IS-7.1.1(A)      0.749              4               |
|         16.66 C          -1.4         31.0            0.00               |
|*                                                                        *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
     49 ST   ISMB250                 (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 83 Check 1

```
|-------------------------------------------------------------------------|
|                                                    Y        PROPERTIES   |
|*************                                       |        IN  CM  UNIT |
|            *  |==============================|   ==| |==     ------------ |
|MEMBER  83  *  |  INDIAN SECTIONS             |     | |       AX =  117.4  |
|            *  | D   ISMB300                  |     | |  --Z  AY =   46.2  |
|DESIGN CODE *  |                              |     | |       AZ =   48.9  |
|  IS-800    *  ================================   ==| |==     SY =  480.3  |
|            *                                                  SZ = 1198.1  |
|            *  |<---LENGTH (ME=    10.00 --->|                 RY =    7.6  |
|*************                                                  RZ =   12.4  |
|                                                                         |
|                  31.7( KN-METR)                                         |
|PARAMETER         |L4                                          STRESSES    |
|IN   NEWT  MM     |                                            IN   NEWT  MM|
|----------------  +                                            -------------|
| KL/R-Y=  132.1   |                                            FA  =   55.4 |
| KL/R-Z=   80.8   +   L4                                  L4   fa  =    1.6 |
| UNL   =10000.0   |                                            FCZ =  109.1 |
| C     =  180.0   +                                            FTZ =  165.0 |
| CMY   =   1.00   |       L4      L5  L4  L4          L5       FCY =  165.0 |
| CMZ   =   0.83   +           L5                               FTY =  165.0 |
| FYLD  =  250.0   |                           L4  L5           fbz =   26.5 |
| NSF   =    1.0   +---+---+---+---+---+---+---+---+---+---|   fby =    0.1 |
| DFF   =    0.0    2.7                                       FV  =  100.0 |
| dff   =    0.0               ABSOLUTE MZ ENVELOPE           fv  =    3.3 |
|                                 (WITH LOAD NO.)                         |
|                                                                         |
|                      MAX FORCE/ MOMENT SUMMARY ( KN-METR)               |
|                      -------------------------                          |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE       0.0          0.0        0.0         0.0         0.0    |
|   LOCATION       0.0          0.0        0.0         0.0         0.0    |
|    LOADING         0            0          0           0           0    |
|                                                                         |
|*************************************************************************|
|*                                                                        *|
|*                         DESIGN SUMMARY ( KN-METR)                      *|
|*                         --------------                                 *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY              MZ           LOCATION            |
|        ========================================================        |
|          PASS        IS-7.1.1(A)      0.273              4              |
|         19.21 C          -0.1         31.7            0.00              |
|*                                                                        *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
     83 D    ISMB300                 (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 84 Check 1

```
|-------------------------------------------------------------------------|
|                                                    Y        PROPERTIES   |
|*************                                       |        IN  CM  UNIT |
|            *  |==============================|   ==| |==     ------------ |
|MEMBER  84  *  |  INDIAN SECTIONS             |     | |       AX =  117.4  |
|            *  | D   ISMB300                  |     | |  --Z  AY =   46.2  |
|DESIGN CODE *  |                              |     | |       AZ =   48.9  |
|  IS-800    *  ================================   ==| |==     SY =  480.3  |
|            *                                                  SZ = 1198.1  |
|            *  |<---LENGTH (ME=    10.00 --->|                 RY =    7.6  |
|*************                                                  RZ =   12.4  |
|                                                                         |
|                  31.7( KN-METR)                                         |
|PARAMETER         |                                        L4  STRESSES    |
|IN   NEWT  MM     |                                            IN   NEWT  MM|
|----------------  +                                            -------------|
| KL/R-Y=  132.1   |                                            FA  =   55.4 |
| KL/R-Z=   80.8   +L4                                  L4      fa  =    1.6 |
| UNL   =10000.0   |                                            FCZ =  109.1 |
| C     =  180.0   +                                            FTZ =  165.0 |
| CMY   =   1.00   |   L5          L4  L4  L5      L4           FCY =  165.0 |
| CMZ   =   0.83   +                          L5                FTY =  165.0 |
| FYLD  =  250.0   |       L5  L4                               fbz =   26.5 |
| NSF   =    1.0   +---+---+---+---+---+---+---+---+---+---|   fby =    0.1 |
| DFF   =    0.0    2.7                                      FV  =  100.0 |
| dff   =    0.0              ABSOLUTE MZ ENVELOPE            fv  =    3.3 |
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY ( KN-METR)                |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE       0.0          0.0        0.0         0.0         0.0    |
|   LOCATION       0.0          0.0        0.0         0.0         0.0    |
|    LOADING         0            0          0           0           0    |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                         DESIGN SUMMARY ( KN-METR)                     *|
|*                         --------------                                *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY              MZ           LOCATION           |
|         ========================================================       |
|          PASS        IS-7.1.1(A)      0.273              4              |
|         19.21 C          -0.1         31.7            10.00             |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
     84 D    ISMB300                 (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 86 Check 1

```
|-------------------------------------------------------------------------|
|                                                    Y        PROPERTIES   |
|*************                                       |        IN  CM  UNIT |
|            *  |==============================|   ===|===     ------------ |
|MEMBER  86  *  |  INDIAN SECTIONS             |      |        AX =   58.4  |
|            *  |     TAP ERED                 |      |   --Z  AY =   35.0  |
|DESIGN CODE *  |                              |      |        AZ =   16.0  |
|  IS-800    *  ================================   ===|===     SY =   80.1  |
|            *                                                  SZ = 1213.4  |
|            *  |<---LENGTH (ME=     2.25 --->|                 RY =    3.7  |
|*************                                                  RZ =   27.0  |
|                                                                         |
|                  0.0( KN-METR)                                          |
|PARAMETER         |                                        L5  STRESSES     |
|IN   NEWT  MM     |                                    L5      IN   NEWT  MM|
|---------------   +                              L5              ------------|
| KL/R-Y=   60.8   |                         L5                   FA  =  150.5 |
| KL/R-Z=    8.3   +                    L5                        fa  =   35.4 |
| UNL   = 2250.0   |               L5                             FCZ =  189.2 |
| C     =  180.0   +                                              FTZ =  224.4 |
| CMY   =   0.60   |       L5  L5                                 FCY =  224.4 |
| CMZ   =   0.60   +   L5                                         FTY =  224.4 |
| FYLD  =  340.0   |L0                                            fbz =    0.0 |
| NSF   =    1.0   +---+---+---+---+---+---+---+---+---+---+---|  fby =    0.0 |
| DFF   =    0.0   -0.0                                           FV  =  136.0 |
| dff   =    0.0              ABSOLUTE MZ ENVELOPE                fv  =    0.0 |
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY ( KN-METR)                |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE       0.0          0.0        0.0         0.0         0.0    |
|   LOCATION       0.0          0.0        0.0         0.0         0.0    |
|    LOADING         0            0          0           0           0    |
|                                                                         |
|*************************************************************************|
|*                                                                        *|
|*                         DESIGN SUMMARY ( KN-METR)                      *|
|*                         --------------                                 *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY              MZ           LOCATION            |
|        ========================================================        |
|          PASS        IS-7.1.1(A)      0.235             4               |
|        206.61 C           0.0          0.0            0.00              |
|*                                                                        *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
     86      TAP ERED                (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 87 Check 1

```
|-------------------------------------------------------------------------|
|                                                    Y        PROPERTIES   |
|*************                                       |        IN  CM  UNIT |
|            *  |==============================|   ===|===     ------------ |
|MEMBER  87  *  |  INDIAN SECTIONS             |      |        AX =   47.6  |
|            *  | ST  ISMB250                  |      |   --Z  AY =   17.2  |
|DESIGN CODE *  |                              |      |        AZ =   20.8  |
|  IS-800    *  ================================   ===|===     SY =   53.6  |
|            *                                                  SZ =  410.6  |
|            *  |<---LENGTH (ME=     2.25 --->|                 RY =    2.7  |
|*************                                                  RZ =   10.4  |
|                                                                         |
|                  19.0( KN-METR)                                         |
|PARAMETER         |                                        L4  STRESSES    |
|IN   NEWT  MM     |                                  L4  L4      IN   NEWT  MM|
|----------------  +                              L4                ------------|
| KL/R-Y=   84.8   |                         L4                     FA  =   93.7 |
| KL/R-Z=   86.7   +                     L4                         fa  =    1.4 |
| UNL   = 2250.0   |                L4                             FCZ =  137.7 |
| C     =  180.0   +                                                FTZ =  165.0 |
| CMY   =   0.88   |            L4                                  FCY =  165.0 |
| CMZ   =   0.67   +   L4  L4                                       FTY =  165.0 |
| FYLD  =  250.0   |L4                                              fbz =   46.2 |
| NSF   =    1.0   +---+---+---+---+---+---+---+---+---+---|        fby =   44.7 |
| DFF   =    0.0    2.4                                            FV  =  100.0 |
| dff   =    0.0               ABSOLUTE MZ ENVELOPE                 fv  =    4.0 |
|                                 (WITH LOAD NO.)                         |
|                                                                         |
|                      MAX FORCE/ MOMENT SUMMARY ( KN-METR)               |
|                      -------------------------                          |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z    |
|                                                                         |
|      VALUE       0.0          0.0        0.0         0.0         0.0    |
|   LOCATION       0.0          0.0        0.0         0.0         0.0    |
|    LOADING         0            0          0           0           0    |
|                                                                         |
|*************************************************************************|
|*                                                                        *|
|*                         DESIGN SUMMARY ( KN-METR)                      *|
|*                         --------------                                 *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY              MZ           LOCATION            |
|         ========================================================        |
|           PASS        IS-7.1.1(A)      0.621             4              |
|           6.73 C          -2.4         19.0           2.25              |
|*                                                                        *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
     87 ST   ISMB250                 (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 88 Check 1

```
|-------------------------------------------------------------------------|
|                                                  Y        PROPERTIES    |
|*************                                     |        IN  CM  UNIT  |
|            *  |==============================|   ===|===     ------------ |
|MEMBER  88  *  |  INDIAN SECTIONS             |      |        AX =   47.6  |
|            *  | ST  ISMB250                  |      |   --Z  AY =   17.2  |
|DESIGN CODE *  |                              |      |        AZ =   20.8  |
|  IS-800    *  ================================   ===|===     SY =   53.6  |
|            *                                                  SZ =  410.6  |
|            *  |<---LENGTH (ME=     2.25 --->|                 RY =    2.7  |
|*************                                                  RZ =   10.4  |
|                                                                         |
|                 18.8( KN-METR)                                          |
|PARAMETER        |L4                                           STRESSES    |
|IN   NEWT  MM    |   L4                                        IN   NEWT  MM|
|---------------  +       L4  L4                                ------------|
| KL/R-Y=   84.8  |               L4                            FA  =  150.0 |
| KL/R-Z=   86.7  +                   L4                      fa  =    6.5 |
| UNL   = 2250.0  |                       L4                    FCZ =  137.7 |
| C     =  180.0  +                                             FTZ =  165.0 |
| CMY   =   0.84  |                           L4  L4            FCY =  165.0 |
| CMZ   =   1.00  +                                   L4        FTY =  165.0 |
| FYLD  =  250.0  |                                       L4    fbz =   45.2 |
| NSF   =    1.0  +---+---+---+---+---+---+---+---+---+---|    fby =   63.2 |
| DFF   =    0.0  18.5                                        FV  =  100.0 |
| dff   =    0.0              ABSOLUTE MZ ENVELOPE            fv  =    4.3 |
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY ( KN-METR)                |
|                     -------------------------                           |
|                                                                         |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z    |
|                                                                         |
|      VALUE       0.0         0.0        0.0         0.0         0.0     |
|   LOCATION       0.0         0.0        0.0         0.0         0.0     |
|    LOADING         0           0          0           0           0     |
|                                                                         |
|*************************************************************************|
|*                                                                        *|
|*                         DESIGN SUMMARY ( KN-METR)                      *|
|*                         --------------                                 *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY              MZ           LOCATION            |
|         ========================================================        |
|          PASS        7.1.2 BEND C     0.711              4              |
|         30.72 T          -3.4         18.6            2.25              |
|*                                                                        *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
     88 ST   ISMB250                 (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 89 Check 1

```
|-------------------------------------------------------------------------|
|                                                  Y        PROPERTIES    |
|*************                                     |        IN  CM  UNIT  |
|            *  |==============================|   ===|===     ------------ |
|MEMBER  89  *  |  INDIAN SECTIONS             |      |        AX =   47.6  |
|            *  | ST  ISMB250                  |      |   --Z  AY =   17.2  |
|DESIGN CODE *  |                              |      |        AZ =   20.8  |
|  IS-800    *  ================================   ===|===     SY =   53.6  |
|            *                                                  SZ =  410.6  |
|            *  |<---LENGTH (ME=     2.25 --->|                 RY =    2.7  |
|*************                                                  RZ =   10.4  |
|                                                                         |
|                 18.9( KN-METR)                                          |
|PARAMETER        |L4                                           STRESSES    |
|IN   NEWT  MM    |   L4                                        IN   NEWT  MM|
|---------------  +       L4  L4                                ------------|
| KL/R-Y=   84.8  |               L4                            FA  =  150.0 |
| KL/R-Z=   86.7  +                                             fa  =   19.4 |
| UNL   = 2250.0  |                   L4  L4                    FCZ =  137.7 |
| C     =  180.0  +                                             FTZ =  165.0 |
| CMY   =   0.72  |                           L4                FCY =  165.0 |
| CMZ   =   0.61  +                                   L4        FTY =  165.0 |
| FYLD  =  250.0  |                                       L4  fbz =   46.2 |
| NSF   =    1.0  +---+---+---+---+---+---+---+---+---+---|   fby =   57.9 |
| DFF   =    0.0  -0.6                                        FV  =  100.0 |
| dff   =    0.0              ABSOLUTE MZ ENVELOPE            fv  =    4.8 |
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY ( KN-METR)                |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE       0.0         0.0        0.0         0.0         0.0     |
|   LOCATION       0.0         0.0        0.0         0.0         0.0     |
|    LOADING         0           0          0           0           0     |
|                                                                         |
|*************************************************************************|
|*                                                                        *|
|*                         DESIGN SUMMARY ( KN-METR)                      *|
|*                         --------------                                 *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY              MZ           LOCATION            |
|         ========================================================        |
|          PASS        IS-7.1.2         0.760              4               |
|         92.18 T          -3.1         18.9            0.00               |
|*                                                                        *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
     89 ST   ISMB250                 (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 90 Check 1

```
|-------------------------------------------------------------------------|
|                                                  Y        PROPERTIES    |
|*************                                     |        IN  CM  UNIT  |
|            *  |==============================|   ===|===     ------------ |
|MEMBER  90  *  |  INDIAN SECTIONS             |      |        AX =   58.4  |
|            *  |     TAP ERED                 |      |   --Z  AY =   35.0  |
|DESIGN CODE *  |                              |      |        AZ =   16.0  |
|  IS-800    *  ================================   ===|===     SY =   80.1  |
|            *                                                  SZ = 1213.4  |
|            *  |<---LENGTH (ME=     2.25 --->|                 RY =    3.7  |
|*************                                                  RZ =   27.0  |
|                                                                         |
|                  1.3( KN-METR)                                          |
|PARAMETER         |                                        L4  STRESSES    |
|IN   NEWT  MM     |                                    L4      IN   NEWT  MM|
|---------------   +                            L4  L4          -------------|
| KL/R-Y=   60.8   |                        L4                  FA  =  150.5 |
| KL/R-Z=    8.3   +                                            fa  =   19.6 |
| UNL   = 2250.0   |               L4  L4                       FCZ =  189.2 |
| C     =  180.0   +                                            FTZ =  224.4 |
| CMY   =   0.40   |           L4                               FCY =  224.4 |
| CMZ   =   0.98   +   L4  L4                                   FTY =  224.4 |
| FYLD  =  340.0   |L4                                          fbz =    1.1 |
| NSF   =    1.0   +---+---+---+---+---+---+---+---+---+---|   fby =    8.0 |
| DFF   =    0.0    1.3                                        FV  =  136.0 |
| dff   =    0.0              ABSOLUTE MZ ENVELOPE             fv  =    0.3 |
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY ( KN-METR)                |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE       0.0         0.0        0.0         0.0         0.0     |
|   LOCATION       0.0         0.0        0.0         0.0         0.0     |
|    LOADING         0           0          0           0           0     |
|                                                                         |
|*************************************************************************|
|*                                                                        *|
|*                         DESIGN SUMMARY ( KN-METR)                      *|
|*                         --------------                                 *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY              MZ           LOCATION            |
|        ========================================================        |
|          PASS        IS-7.1.1(A)      0.172              4              |
|        114.65 C          -0.6         -1.3            0.00              |
|*                                                                        *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
     90      TAP ERED                (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 91 Check 1

```
|-------------------------------------------------------------------------|
|                                                    Y        PROPERTIES   |
|*************                                       |        IN  CM  UNIT |
|            *  |==============================|    |== ==|     ------------ |
|MEMBER  91  *  |  INDIAN SECTIONS             |    |     |     AX =   18.2  |
|            *  | FR  ISMC75                   |    |     |--Z  AY =    7.2  |
|DESIGN CODE *  |                              |    |     |     AZ =    8.0  |
|  IS-800    *  ================================   |== ==|     SY =   39.1  |
|            *                                                   SZ =   41.6  |
|            *  |<---LENGTH (ME=     1.66 --->|                  RY =    2.9  |
|*************                                                   RZ =    2.9  |
|                                                                         |
|                  0.4( KN-METR)                                          |
|PARAMETER         |                                          L4  STRESSES     |
|IN   NEWT  MM     |                                              IN   NEWT  MM|
|----------------  +                                     L4       -------------|
| KL/R-Y=   56.5   |                                  L4          FA  =  125.6 |
| KL/R-Z=   56.5   +                                              fa  =   35.8 |
| UNL   = 1655.3   |L4                         L4                FCZ =  156.8 |
| C     =  400.0   +   L4                   L4                  FTZ =  165.0 |
| CMY   =   0.82   |       L4                                    FCY =  165.0 |
| CMZ   =   0.40   +           L4                                 FTY =  165.0 |
| FYLD  =  250.0   |               L4  L4                        fbz =   10.8 |
| NSF   =    1.0   +---+---+---+---+---+---+---+---+---+---|    fby =   16.0 |
| DFF   =    0.0    0.0                                         FV  =  100.0 |
| dff   =    0.0               ABSOLUTE MZ ENVELOPE             fv  =    0.7 |
|                                 (WITH LOAD NO.)                         |
|                                                                         |
|                      MAX FORCE/ MOMENT SUMMARY ( KN-METR)               |
|                      -------------------------                          |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z    |
|                                                                         |
|      VALUE       0.0          0.0        0.0         0.0         0.0    |
|   LOCATION       0.0          0.0        0.0         0.0         0.0    |
|    LOADING         0            0          0           0           0    |
|                                                                         |
|*************************************************************************|
|*                                                                        *|
|*                         DESIGN SUMMARY ( KN-METR)                      *|
|*                         --------------                                 *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY              MZ           LOCATION            |
|         ========================================================        |
|          PASS        IS-7.1.1(B)      0.404             4               |
|         65.20 C           0.6          0.4            1.66              |
|*                                                                        *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
     91 FR   ISMC75                  (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 92 Check 1

```
|-------------------------------------------------------------------------|
|                                                  Y        PROPERTIES    |
|*************                                     |        IN  CM  UNIT  |
|            *  |==============================|   ===|===     ------------ |
|MEMBER  92  *  |  INDIAN SECTIONS             |      |        AX =   58.4  |
|            *  |     TAP ERED                 |      |   --Z  AY =   35.0  |
|DESIGN CODE *  |                              |      |        AZ =   16.0  |
|  IS-800    *  ================================   ===|===     SY =   80.1  |
|            *                                                  SZ = 1213.4  |
|            *  |<---LENGTH (ME=     2.25 --->|                 RY =    3.7  |
|*************                                                  RZ =   27.0  |
|                                                                         |
|                  1.6( KN-METR)                                          |
|PARAMETER         |L4                                          STRESSES    |
|IN   NEWT  MM     |   L4                                       IN   NEWT  MM|
|---------------   +       L4  L4                               -------------|
| KL/R-Y=   60.8   |               L4                           FA  =  150.5 |
| KL/R-Z=    8.3   +                   L4                      fa  =    8.6 |
| UNL   = 2250.0   |                       L4                  FCZ =  189.2 |
| C     =  180.0   +                                           FTZ =  224.4 |
| CMY   =   0.51   |                           L4  L4           FCY =  224.4 |
| CMZ   =   0.89   +                                   L4      FTY =  224.4 |
| FYLD  =  340.0   |                                      L4   fbz =    1.0 |
| NSF   =    1.0   +---+---+---+---+---+---+---+---+---+---|   fby =    7.9 |
| DFF   =    0.0    1.1                                       FV  =  136.0 |
| dff   =    0.0              ABSOLUTE MZ ENVELOPE            fv  =    0.2 |
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY ( KN-METR)                |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE       0.0          0.0        0.0         0.0         0.0    |
|   LOCATION       0.0          0.0        0.0         0.0         0.0    |
|    LOADING         0            0          0           0           0    |
|                                                                         |
|*************************************************************************|
|*                                                                        *|
|*                         DESIGN SUMMARY ( KN-METR)                      *|
|*                         --------------                                 *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY              MZ           LOCATION            |
|         ========================================================        |
|          PASS        IS-7.1.1(A)      0.097             4               |
|         50.40 C           0.6         -1.2            2.25              |
|*                                                                        *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
     92      TAP ERED                (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 93 Check 1

```
|-------------------------------------------------------------------------|
|                                                    Y        PROPERTIES   |
|*************                                       |        IN  CM  UNIT |
|            *  |==============================|    |== ==|     ------------ |
|MEMBER  93  *  |  INDIAN SECTIONS             |    |     |     AX =   18.2  |
|            *  | FR  ISMC75                   |    |     |--Z  AY =    7.2  |
|DESIGN CODE *  |                              |    |     |     AZ =    8.0  |
|  IS-800    *  ================================   |== ==|     SY =   39.1  |
|            *                                                   SZ =   41.6  |
|            *  |<---LENGTH (ME=     1.66 --->|                  RY =    2.9  |
|*************                                                   RZ =    2.9  |
|                                                                         |
|                  0.5( KN-METR)                                          |
|PARAMETER         |                                          L4  STRESSES     |
|IN   NEWT  MM     |                                              IN   NEWT  MM|
|----------------  +L4                                      L4      -------------|
| KL/R-Y=   56.5   |   L4                              L4           FA  =  125.6 |
| KL/R-Z=   56.5   +       L4                                      fa  =   24.8 |
| UNL   = 1655.3   |                               L4             FCZ =  156.8 |
| C     =  400.0   +           L4          L4                     FTZ =  165.0 |
| CMY   =   0.52   |               L4                            FCY =  165.0 |
| CMZ   =   0.40   +                                             FTY =  165.0 |
| FYLD  =  250.0   |                   L1                        fbz =   12.6 |
| NSF   =    1.0   +---+---+---+---+---+---+---+---+---+---|    fby =    0.3 |
| DFF   =    0.0   -0.0                                          FV  =  100.0 |
| dff   =    0.0              ABSOLUTE MZ ENVELOPE               fv  =    0.9 |
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY ( KN-METR)                |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE       0.0          0.0        0.0         0.0         0.0    |
|   LOCATION       0.0          0.0        0.0         0.0         0.0    |
|    LOADING         0            0          0           0           0    |
|                                                                         |
|*************************************************************************|
|*                                                                        *|
|*                         DESIGN SUMMARY ( KN-METR)                      *|
|*                         --------------                                 *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY              MZ           LOCATION            |
|        ========================================================        |
|          PASS        IS-7.1.1(B)      0.247              4              |
|         45.19 C           0.0          0.5             1.66             |
|*                                                                        *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
     93 FR   ISMC75                  (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 94 Check 1

```
|-------------------------------------------------------------------------|
|                                                    Y        PROPERTIES   |
|*************                                       |        IN  CM  UNIT |
|            *  |==============================|   ===|===     ------------ |
|MEMBER  94  *  |  INDIAN SECTIONS             |      |        AX =   58.4  |
|            *  |     TAP ERED                 |      |   --Z  AY =   35.0  |
|DESIGN CODE *  |                              |      |        AZ =   16.0  |
|  IS-800    *  ================================   ===|===     SY =   80.1  |
|            *                                                  SZ = 1213.4  |
|            *  |<---LENGTH (ME=     2.25 --->|                 RY =    3.7  |
|*************                                                  RZ =   27.0  |
|                                                                         |
|                  0.9( KN-METR)                                          |
|PARAMETER         |L4                                          STRESSES    |
|IN   NEWT  MM     |   L4                                       IN   NEWT  MM|
|---------------   +       L4  L4                               -------------|
| KL/R-Y=   60.8   |               L4                           FA  =  150.5 |
| KL/R-Z=    8.3   +                                            fa  =    1.8 |
| UNL   = 2250.0   |                   L4  L4                   FCZ =  189.2 |
| C     =  180.0   +                                            FTZ =  224.4 |
| CMY   =   0.45   |                           L4               FCY =  224.4 |
| CMZ   =   0.81   +                               L4  L4       FTY =  224.4 |
| FYLD  =  340.0   |                                     L4     fbz =    0.4 |
| NSF   =    1.0   +---+---+---+---+---+---+---+---+---+---|    fby =    2.2 |
| DFF   =    0.0    0.4                                        FV  =  136.0 |
| dff   =    0.0              ABSOLUTE MZ ENVELOPE             fv  =    0.1 |
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY ( KN-METR)                |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE       0.0          0.0        0.0         0.0         0.0    |
|   LOCATION       0.0          0.0        0.0         0.0         0.0    |
|    LOADING         0            0          0           0           0    |
|                                                                         |
|*************************************************************************|
|*                                                                        *|
|*                         DESIGN SUMMARY ( KN-METR)                      *|
|*                         --------------                                 *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY              MZ           LOCATION            |
|        ========================================================        |
|          PASS        IS-7.1.1(A)      0.024              4              |
|         10.56 C          -0.2         -0.5            2.25              |
|*                                                                        *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
     94      TAP ERED                (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 95 Check 1

```
|-------------------------------------------------------------------------|
|                                                    Y        PROPERTIES   |
|*************                                       |        IN  CM  UNIT |
|            *  |==============================|    |== ==|     ------------ |
|MEMBER  95  *  |  INDIAN SECTIONS             |    |     |     AX =   18.2  |
|            *  | FR  ISMC75                   |    |     |--Z  AY =    7.2  |
|DESIGN CODE *  |                              |    |     |     AZ =    8.0  |
|  IS-800    *  ================================    |== ==|     SY =   39.1  |
|            *                                                   SZ =   41.6  |
|            *  |<---LENGTH (ME=     1.66 --->|                  RY =    2.9  |
|*************                                                   RZ =    2.9  |
|                                                                         |
|                   0.5( KN-METR)                                         |
|PARAMETER         |L4                                          STRESSES    |
|IN   NEWT  MM     |   L4                                       IN   NEWT  MM|
|----------------  +       L4                                L4  -------------|
| KL/R-Y=   56.5   |                                            FA  =  125.6 |
| KL/R-Z=   56.5   +           L4                        L4      fa  =   15.6 |
| UNL   = 1655.3   |               L4                           FCZ =  156.8 |
| C     =  180.0   +                                 L4          FTZ =  165.0 |
| CMY   =   1.00   |                            L4              FCY =  165.0 |
| CMZ   =   0.40   +                   L4                       FTY =  165.0 |
| FYLD  =  250.0   |                       L4                   fbz =   11.0 |
| NSF   =    1.0   +---+---+---+---+---+---+---+---+---+---|   fby =    8.4 |
| DFF   =    0.0    0.0                                        FV  =  100.0 |
| dff   =    0.0              ABSOLUTE MZ ENVELOPE             fv  =    0.5 |
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY ( KN-METR)                |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE       0.0          0.0        0.0         0.0         0.0    |
|   LOCATION       0.0          0.0        0.0         0.0         0.0    |
|    LOADING         0            0          0           0           0    |
|                                                                         |
|*************************************************************************|
|*                                                                        *|
|*                         DESIGN SUMMARY ( KN-METR)                      *|
|*                         --------------                                 *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY              MZ           LOCATION            |
|        ========================================================        |
|          PASS        IS-7.1.1(A)      0.246              4              |
|         28.36 C          -0.3         -0.5            0.00              |
|*                                                                        *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
     95 FR   ISMC75                  (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 96 Check 1

```
|-------------------------------------------------------------------------|
|                                                  Y        PROPERTIES    |
|*************                                     |        IN  CM  UNIT  |
|            *  |==============================|   ===|===     ------------ |
|MEMBER  96  *  |  INDIAN SECTIONS             |      |        AX =   58.4  |
|            *  |     TAP ERED                 |      |   --Z  AY =   35.0  |
|DESIGN CODE *  |                              |      |        AZ =   16.0  |
|  IS-800    *  ================================   ===|===     SY =   80.1  |
|            *                                                  SZ = 1213.4  |
|            *  |<---LENGTH (ME=     0.40 --->|                 RY =    3.7  |
|*************                                                  RZ =   27.0  |
|                                                                         |
|                  0.0( KN-METR)                                          |
|PARAMETER         |L4                                          STRESSES    |
|IN   NEWT  MM     |   L4  L4                                   IN   NEWT  MM|
|---------------   +           L4                               ------------|
| KL/R-Y=   10.8   |               L4                           FA  =  203.4 |
| KL/R-Z=    1.5   +                                            fa  =    0.0 |
| UNL   =  400.0   |                   L4  L4                   FCZ =  224.0 |
| C     =  180.0   +                                            FTZ =  224.4 |
| CMY   =    0.60  |                           L4  L4           FCY =  224.4 |
| CMZ   =    0.60  +                                 L4         FTY =  224.4 |
| FYLD  =  340.0   |                                    L0      fbz =    0.0 |
| NSF   =    1.0   +---+---+---+---+---+---+---+---+---+---|    fby =    0.0 |
| DFF   =    0.0   -0.0                                         FV  =  136.0 |
| dff   =    0.0              ABSOLUTE MZ ENVELOPE              fv  =    0.0 |
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY ( KN-METR)                |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE       0.0         0.0        0.0         0.0         0.0     |
|   LOCATION       0.0         0.0        0.0         0.0         0.0     |
|    LOADING         0           0          0           0           0     |
|                                                                         |
|*************************************************************************|
|*                                                                        *|
|*                         DESIGN SUMMARY ( KN-METR)                      *|
|*                         --------------                                 *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY              MZ           LOCATION            |
|         ========================================================        |
|           PASS        IS-7.1.1(A)      0.000             1              |
|           0.18 C           0.0         -0.0            0.00             |
|*                                                                        *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
     96      TAP ERED                (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 97 Check 1

```
|-------------------------------------------------------------------------|
|                                                    Y        PROPERTIES   |
|*************                                       |        IN  CM  UNIT |
|            *  |==============================|    |== ==|     ------------ |
|MEMBER  97  *  |  INDIAN SECTIONS             |    |     |     AX =   18.2  |
|            *  | FR  ISMC75                   |    |     |--Z  AY =    7.2  |
|DESIGN CODE *  |                              |    |     |     AZ =    8.0  |
|  IS-800    *  ================================    |== ==|     SY =   39.1  |
|            *                                                   SZ =   41.6  |
|            *  |<---LENGTH (ME=     1.66 --->|                  RY =    2.9  |
|*************                                                   RZ =    2.9  |
|                                                                         |
|                   1.6( KN-METR)                                         |
|PARAMETER         |L4                                                    STRESSES     |
|IN   NEWT  MM     |   L4                                                 IN   NEWT  MM|
|---------------   +       L4                                             -------------|
| KL/R-Y=   56.5   |                                                      FA  =  125.6 |
| KL/R-Z=   56.5   +           L4                                         fa  =    4.5 |
| UNL   = 1655.3   |               L4                                    FCZ =  156.8 |
| C     =  400.0   +                                              L4      FTZ =  165.0 |
| CMY   =   0.48   |                    L4              L4               FCY =  165.0 |
| CMZ   =   0.45   +                        L4      L4                    FTY =  165.0 |
| FYLD  =  250.0   |                             L4                      fbz =   38.2 |
| NSF   =    1.0   +---+---+---+---+---+---+---+---+---+---|              fby =    2.2 |
| DFF   =    0.0    0.0                                                   FV  =  100.0 |
| dff   =    0.0              ABSOLUTE MZ ENVELOPE                        fv  =    2.0 |
|                                (WITH LOAD NO.)                                      |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY ( KN-METR)                |
|                     -------------------------                           |
|                                                                         |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z     |
|                                                                         |
|      VALUE       0.0         0.0        0.0         0.0         0.0     |
|   LOCATION       0.0         0.0        0.0         0.0         0.0     |
|    LOADING         0           0          0           0           0     |
|                                                                         |
|*************************************************************************|
|*                                                                        *|
|*                         DESIGN SUMMARY ( KN-METR)                      *|
|*                         --------------                                 *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY              MZ           LOCATION            |
|         ========================================================        |
|           PASS        IS-7.1.1(A)      0.293             4              |
|           8.19 C           0.1          1.6            0.00             |
|*                                                                        *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
     97 FR   ISMC75                  (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 98 Check 1

```
|-------------------------------------------------------------------------|
|                                                    Y        PROPERTIES   |
|*************                                       |        IN  CM  UNIT |
|            *  |==============================|   ===|===     ------------ |
|MEMBER  98  *  |  INDIAN SECTIONS             |      |        AX =   47.6  |
|            *  | ST  ISMB250                  |      |   --Z  AY =   17.2  |
|DESIGN CODE *  |                              |      |        AZ =   20.8  |
|  IS-800    *  ================================   ===|===     SY =   53.6  |
|            *                                                  SZ =  410.6  |
|            *  |<---LENGTH (ME=     2.25 --->|                RY =    2.7  |
|*************                                                  RZ =   10.4  |
|                                                                         |
|                  19.0( KN-METR)                                         |
|PARAMETER         |                                        L4  STRESSES    |
|IN   NEWT  MM     |                                   L4  L4      IN   NEWT  MM|
|----------------  +                               L4  L4           -------------|
| KL/R-Y=   84.8   |                          L4                    FA  =   93.7 |
| KL/R-Z=   86.7   +                                                fa  =    1.4 |
| UNL   = 2250.0   |                L4  L4                         FCZ =  137.7 |
| C     =  180.0   +                                                FTZ =  165.0 |
| CMY   =   0.88   |            L4                                 FCY =  165.0 |
| CMZ   =   0.67   +    L4  L4                                      FTY =  165.0 |
| FYLD  =  250.0   |L4                                             fbz =   46.2 |
| NSF   =    1.0   +---+---+---+---+---+---+---+---+---+---+---|   fby =   44.7 |
| DFF   =    0.0    2.4                                           FV  =  100.0 |
| dff   =    0.0                ABSOLUTE MZ ENVELOPE              fv  =    4.0 |
|                                  (WITH LOAD NO.)                        |
|                                                                         |
|                       MAX FORCE/ MOMENT SUMMARY ( KN-METR)              |
|                       -------------------------                         |
|                                                                         |
|                 AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z  |
|                                                                         |
|      VALUE        0.0         0.0        0.0         0.0         0.0   |
|   LOCATION        0.0         0.0        0.0         0.0         0.0   |
|    LOADING          0           0          0           0           0     |
|                                                                         |
|*************************************************************************|
|*                                                                        *|
|*                         DESIGN SUMMARY ( KN-METR)                      *|
|*                         --------------                                 *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY              MZ           LOCATION            |
|         ========================================================        |
|           PASS        IS-7.1.1(A)      0.621             4              |
|           6.73 C          -2.4        -19.0            2.25             |
|*                                                                        *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
     98 ST   ISMB250                 (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 99 Check 1

```
|-------------------------------------------------------------------------|
|                                                  Y        PROPERTIES    |
|*************                                     |        IN  CM  UNIT  |
|            *  |==============================|   ===|===     ------------ |
|MEMBER  99  *  |  INDIAN SECTIONS             |      |        AX =   47.6  |
|            *  | ST  ISMB250                  |      |   --Z  AY =   17.2  |
|DESIGN CODE *  |                              |      |        AZ =   20.8  |
|  IS-800    *  ================================   ===|===     SY =   53.6  |
|            *                                                  SZ =  410.6  |
|            *  |<---LENGTH (ME=     2.25 --->|                 RY =    2.7  |
|*************                                                  RZ =   10.4  |
|                                                                         |
|                 18.8( KN-METR)                                          |
|PARAMETER        |L4                                           STRESSES    |
|IN   NEWT  MM    |   L4  L4                                    IN   NEWT  MM|
|---------------  +       L4  L4                                ------------|
| KL/R-Y=   84.8  |               L4                            FA  =  150.0 |
| KL/R-Z=   86.7  +                   L4                      fa  =    6.5 |
| UNL   = 2250.0  |                       L4                    FCZ =  137.7 |
| C     =  180.0  +                                             FTZ =  165.0 |
| CMY   =   0.84  |                           L4  L4            FCY =  165.0 |
| CMZ   =   1.00  +                                   L4        FTY =  165.0 |
| FYLD  =  250.0  |                                       L4    fbz =   45.2 |
| NSF   =    1.0  +---+---+---+---+---+---+---+---+---+---|    fby =   63.2 |
| DFF   =    0.0  18.5                                        FV  =  100.0 |
| dff   =    0.0              ABSOLUTE MZ ENVELOPE            fv  =    4.3 |
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY ( KN-METR)                |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE       0.0          0.0        0.0         0.0         0.0    |
|   LOCATION       0.0          0.0        0.0         0.0         0.0    |
|    LOADING         0            0          0           0           0    |
|                                                                         |
|*************************************************************************|
|*                                                                        *|
|*                         DESIGN SUMMARY ( KN-METR)                      *|
|*                         --------------                                 *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY              MZ           LOCATION            |
|         ========================================================        |
|          PASS        7.1.2 BEND C     0.711              4              |
|         30.72 T          -3.4        -18.6            2.25              |
|*                                                                        *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
     99 ST   ISMB250                 (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 100 Check 1

```
|-------------------------------------------------------------------------|
|                                                  Y        PROPERTIES    |
|*************                                     |        IN  CM  UNIT  |
|            *  |==============================|   ===|===     ------------ |
|MEMBER 100  *  |  INDIAN SECTIONS             |      |        AX =   47.6  |
|            *  | ST  ISMB250                  |      |   --Z  AY =   17.2  |
|DESIGN CODE *  |                              |      |        AZ =   20.8  |
|  IS-800    *  ================================   ===|===     SY =   53.6  |
|            *                                                  SZ =  410.6  |
|            *  |<---LENGTH (ME=     2.25 --->|                 RY =    2.7  |
|*************                                                  RZ =   10.4  |
|                                                                         |
|                 18.9( KN-METR)                                          |
|PARAMETER        |L4                                           STRESSES    |
|IN   NEWT  MM    |   L4                                        IN   NEWT  MM|
|---------------  +           L4                                -------------|
| KL/R-Y=   84.8  |               L4                            FA  =  150.0 |
| KL/R-Z=   86.7  +                   L4                      fa  =   19.4 |
| UNL   = 2250.0  |                       L4                    FCZ =  137.7 |
| C     =  180.0  +                                             FTZ =  165.0 |
| CMY   =   0.72  |                           L4  L4            FCY =  165.0 |
| CMZ   =   0.61  +                                   L4        FTY =  165.0 |
| FYLD  =  250.0  |                                       L4    fbz =   46.2 |
| NSF   =    1.0  +---+---+---+---+---+---+---+---+---+---|    fby =   57.9 |
| DFF   =    0.0  -0.6                                        FV  =  100.0 |
| dff   =    0.0              ABSOLUTE MZ ENVELOPE            fv  =    4.8 |
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY ( KN-METR)                |
|                     -------------------------                           |
|                                                                         |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z    |
|                                                                         |
|      VALUE       0.0         0.0        0.0         0.0         0.0     |
|   LOCATION       0.0         0.0        0.0         0.0         0.0     |
|    LOADING         0           0          0           0           0     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                         DESIGN SUMMARY ( KN-METR)                     *|
|*                         --------------                                *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY              MZ           LOCATION           |
|         ========================================================       |
|          PASS        IS-7.1.2         0.760              4             |
|         92.18 T          -3.1        -18.9            0.00             |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
   100 ST   ISMB250                 (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 101 Check 1

```
|-------------------------------------------------------------------------|
|                                                    Y        PROPERTIES   |
|*************                                       |        IN  CM  UNIT |
|            *  |==============================|   ===|===     ------------ |
|MEMBER 101  *  |  INDIAN SECTIONS             |      |        AX =   58.4  |
|            *  |     TAP ERED                 |      |   --Z  AY =   35.0  |
|DESIGN CODE *  |                              |      |        AZ =   16.0  |
|  IS-800    *  ================================   ===|===     SY =   80.1  |
|            *                                                  SZ = 1213.4  |
|            *  |<---LENGTH (ME=     2.25 --->|                 RY =    3.7  |
|*************                                                  RZ =   27.0  |
|                                                                         |
|                  0.0( KN-METR)                                          |
|PARAMETER         |                                          L5  STRESSES     |
|IN   NEWT  MM     |                                     L5       IN   NEWT  MM|
|---------------   +                              L5  L5           -------------|
| KL/R-Y=   60.8   |                         L5                    FA  =  150.5 |
| KL/R-Z=    8.3   +                     L5                        fa  =   35.4 |
| UNL   = 2250.0   |                L5  L5                         FCZ =  189.2 |
| C     =  180.0   +                                               FTZ =  224.4 |
| CMY   =   0.60   |       L5  L5                                   FCY =  224.4 |
| CMZ   =   0.60   +   L5                                        FTY =  224.4 |
| FYLD  =  340.0   |L0                                              fbz =    0.0 |
| NSF   =    1.0   +---+---+---+---+---+---+---+---+---+---|      fby =    0.0 |
| DFF   =    0.0   -0.0                                          FV  =  136.0 |
| dff   =    0.0               ABSOLUTE MZ ENVELOPE              fv  =    0.0 |
|                                 (WITH LOAD NO.)                         |
|                                                                         |
|                      MAX FORCE/ MOMENT SUMMARY ( KN-METR)               |
|                      -------------------------                          |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z     |
|                                                                         |
|      VALUE       0.0         0.0        0.0         0.0         0.0     |
|   LOCATION       0.0         0.0        0.0         0.0         0.0     |
|    LOADING         0           0          0           0           0     |
|                                                                         |
|*************************************************************************|
|*                                                                        *|
|*                         DESIGN SUMMARY ( KN-METR)                      *|
|*                         --------------                                 *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY              MZ           LOCATION            |
|        ========================================================        |
|          PASS        IS-7.1.1(A)      0.235              4              |
|        206.61 C           0.0          0.0            0.00              |
|*                                                                        *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
   101      TAP ERED                (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 102 Check 1

```
|-------------------------------------------------------------------------|
|                                                    Y        PROPERTIES   |
|*************                                       |        IN  CM  UNIT |
|            *  |==============================|   ===|===     ------------ |
|MEMBER 102  *  |  INDIAN SECTIONS             |      |        AX =   58.4  |
|            *  |     TAP ERED                 |      |   --Z  AY =   35.0  |
|DESIGN CODE *  |                              |      |        AZ =   16.0  |
|  IS-800    *  ================================   ===|===     SY =   80.1  |
|            *                                                  SZ = 1213.4  |
|            *  |<---LENGTH (ME=     2.25 --->|                 RY =    3.7  |
|*************                                                  RZ =   27.0  |
|                                                                         |
|                  1.3( KN-METR)                                          |
|PARAMETER         |                                        L4  STRESSES     |
|IN   NEWT  MM     |                                    L4      IN   NEWT  MM|
|---------------   +                            L4                ----------|
| KL/R-Y=   60.8   |                         L4                   FA  =  150.5 |
| KL/R-Z=    8.3   +                                              fa  =   19.6 |
| UNL   = 2250.0   |               L4                             FCZ =  189.2 |
| C     =  180.0   +                                              FTZ =  224.4 |
| CMY   =   0.40   |           L4                                 FCY =  224.4 |
| CMZ   =   0.98   +   L4                                          FTY =  224.4 |
| FYLD  =  340.0   |L4                                            fbz =    1.1 |
| NSF   =    1.0   +---+---+---+---+---+---+---+---+---+---|      fby =    8.0 |
| DFF   =    0.0    1.3                                        FV  =  136.0 |
| dff   =    0.0               ABSOLUTE MZ ENVELOPE            fv  =    0.3 |
|                                 (WITH LOAD NO.)                         |
|                                                                         |
|                      MAX FORCE/ MOMENT SUMMARY ( KN-METR)               |
|                      -------------------------                          |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z    |
|                                                                         |
|      VALUE       0.0          0.0        0.0         0.0         0.0    |
|   LOCATION       0.0          0.0        0.0         0.0         0.0    |
|    LOADING         0            0          0           0           0    |
|                                                                         |
|*************************************************************************|
|*                                                                        *|
|*                         DESIGN SUMMARY ( KN-METR)                      *|
|*                         --------------                                 *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY              MZ           LOCATION            |
|        ========================================================        |
|          PASS        IS-7.1.1(A)      0.172              4              |
|        114.65 C          -0.6          1.3             0.00              |
|*                                                                        *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
   102      TAP ERED                (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 103 Check 1

```
|-------------------------------------------------------------------------|
|                                                    Y        PROPERTIES   |
|*************                                       |        IN  CM  UNIT |
|            *  |==============================|    |== ==|     ------------ |
|MEMBER 103  *  |  INDIAN SECTIONS             |    |     |     AX =   18.2  |
|            *  | FR  ISMC75                   |    |     |--Z  AY =    7.2  |
|DESIGN CODE *  |                              |    |     |     AZ =    8.0  |
|  IS-800    *  ================================   |== ==|     SY =   39.1  |
|            *                                                   SZ =   41.6  |
|            *  |<---LENGTH (ME=     1.66 --->|                 RY =    2.9  |
|*************                                                   RZ =    2.9  |
|                                                                         |
|                  0.4( KN-METR)                                          |
|PARAMETER         |                                          L4  STRESSES     |
|IN   NEWT  MM     |                                              IN   NEWT  MM|
|----------------  +                                     L4       -------------|
| KL/R-Y=   56.5   |                                  L4          FA  =  125.6 |
| KL/R-Z=   56.5   +                                               fa  =   35.8 |
| UNL   = 1655.3   |L4                         L4                 FCZ =  156.8 |
| C     =  400.0   +   L4                   L4                   FTZ =  165.0 |
| CMY   =   0.82   |       L4                                     FCY =  165.0 |
| CMZ   =   0.40   +            L4                               FTY =  165.0 |
| FYLD  =  250.0   |               L4  L4                         fbz =   10.8 |
| NSF   =    1.0   +---+---+---+---+---+---+---+---+---+---+     fby =   16.0 |
| DFF   =    0.0    0.0                                          FV  =  100.0 |
| dff   =    0.0               ABSOLUTE MZ ENVELOPE              fv  =    0.7 |
|                                 (WITH LOAD NO.)                         |
|                                                                         |
|                      MAX FORCE/ MOMENT SUMMARY ( KN-METR)               |
|                      -------------------------                          |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z    |
|                                                                         |
|      VALUE       0.0          0.0        0.0         0.0         0.0    |
|   LOCATION       0.0          0.0        0.0         0.0         0.0    |
|    LOADING         0            0          0           0           0    |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                         DESIGN SUMMARY ( KN-METR)                     *|
|*                         --------------                                *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY              MZ           LOCATION           |
|         ========================================================       |
|          PASS        IS-7.1.1(B)      0.404             4              |
|         65.20 C          -0.6          0.4            1.66              |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
   103 FR   ISMC75                 (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 104 Check 1

```
|-------------------------------------------------------------------------|
|                                                  Y        PROPERTIES    |
|*************                                     |        IN  CM  UNIT  |
|            *  |==============================|   ===|===     ------------ |
|MEMBER 104  *  |  INDIAN SECTIONS             |      |        AX =   58.4  |
|            *  |     TAP ERED                 |      |   --Z  AY =   35.0  |
|DESIGN CODE *  |                              |      |        AZ =   16.0  |
|  IS-800    *  ================================   ===|===     SY =   80.1  |
|            *                                                  SZ = 1213.4  |
|            *  |<---LENGTH (ME=     2.25 --->|                 RY =    3.7  |
|*************                                                  RZ =   27.0  |
|                                                                         |
|                  1.6( KN-METR)                                          |
|PARAMETER         |L4                                          STRESSES    |
|IN   NEWT  MM     |   L4  L4                                   IN   NEWT  MM|
|---------------   +           L4                               -------------|
| KL/R-Y=   60.8   |               L4                           FA  =  150.5 |
| KL/R-Z=    8.3   +                                            fa  =    8.6 |
| UNL   = 2250.0   |                       L4                   FCZ =  189.2 |
| C     =  180.0   +                                            FTZ =  224.4 |
| CMY   =   0.51   |                           L4  L4           FCY =  224.4 |
| CMZ   =   0.89   +                                   L4       FTY =  224.4 |
| FYLD  =  340.0   |                                      L4   fbz =    1.0 |
| NSF   =    1.0   +---+---+---+---+---+---+---+---+---+---|   fby =    7.9 |
| DFF   =    0.0    1.1                                       FV  =  136.0 |
| dff   =    0.0              ABSOLUTE MZ ENVELOPE            fv  =    0.2 |
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY ( KN-METR)                |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE       0.0         0.0        0.0         0.0         0.0     |
|   LOCATION       0.0         0.0        0.0         0.0         0.0     |
|    LOADING         0           0          0           0           0     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                         DESIGN SUMMARY ( KN-METR)                     *|
|*                         --------------                                *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY              MZ           LOCATION           |
|         ========================================================       |
|          PASS        IS-7.1.1(A)      0.097             4               |
|         50.40 C           0.6          1.2            2.25              |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
   104      TAP ERED                (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 105 Check 1

```
|-------------------------------------------------------------------------|
|                                                    Y        PROPERTIES   |
|*************                                       |        IN  CM  UNIT |
|            *  |==============================|    |== ==|     ------------ |
|MEMBER 105  *  |  INDIAN SECTIONS             |    |     |     AX =   18.2  |
|            *  | FR  ISMC75                   |    |     |--Z  AY =    7.2  |
|DESIGN CODE *  |                              |    |     |     AZ =    8.0  |
|  IS-800    *  ================================   |== ==|     SY =   39.1  |
|            *                                                   SZ =   41.6  |
|            *  |<---LENGTH (ME=     1.66 --->|                  RY =    2.9  |
|*************                                                   RZ =    2.9  |
|                                                                         |
|                   0.5( KN-METR)                                         |
|PARAMETER          |                                          L4  STRESSES     |
|IN   NEWT  MM      |                                              IN   NEWT  MM|
|---------------  +L4                                       L4      -------------|
| KL/R-Y=   56.5  |   L4                              L4           FA  =  125.6 |
| KL/R-Z=   56.5  +       L4                                     fa  =   24.8 |
| UNL   = 1655.3  |                              L4              FCZ =  156.8 |
| C     =  400.0  +           L4          L4                     FTZ =  165.0 |
| CMY   =   0.52  |               L4                           FCY =  165.0 |
| CMZ   =   0.40  +                                            FTY =  165.0 |
| FYLD  =  250.0  |                   L1                       fbz =   12.6 |
| NSF   =    1.0  +---+---+---+---+---+---+---+---+---+---|   fby =    0.3 |
| DFF   =    0.0  -0.0                                         FV  =  100.0 |
| dff   =    0.0              ABSOLUTE MZ ENVELOPE             fv  =    0.9 |
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY ( KN-METR)                |
|                     -------------------------                           |
|                                                                         |
|               AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z     |
|                                                                         |
|      VALUE       0.0         0.0        0.0         0.0         0.0     |
|   LOCATION       0.0         0.0        0.0         0.0         0.0     |
|    LOADING         0           0          0           0           0     |
|                                                                         |
|*************************************************************************|
|*                                                                        *|
|*                         DESIGN SUMMARY ( KN-METR)                      *|
|*                         --------------                                 *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY              MZ           LOCATION            |
|         ========================================================        |
|          PASS        IS-7.1.1(B)      0.247             4               |
|         45.19 C          -0.0          0.5            1.66              |
|*                                                                        *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
   105 FR   ISMC75                  (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 106 Check 1

```
|-------------------------------------------------------------------------|
|                                                  Y        PROPERTIES    |
|*************                                     |        IN  CM  UNIT  |
|            *  |==============================|   ===|===     ------------ |
|MEMBER 106  *  |  INDIAN SECTIONS             |      |        AX =   58.4  |
|            *  |     TAP ERED                 |      |   --Z  AY =   35.0  |
|DESIGN CODE *  |                              |      |        AZ =   16.0  |
|  IS-800    *  ================================   ===|===     SY =   80.1  |
|            *                                                  SZ = 1213.4  |
|            *  |<---LENGTH (ME=     2.25 --->|                 RY =    3.7  |
|*************                                                  RZ =   27.0  |
|                                                                         |
|                  0.9( KN-METR)                                          |
|PARAMETER         |L4                                          STRESSES    |
|IN   NEWT  MM     |   L4                                       IN   NEWT  MM|
|---------------   +           L4                               -------------|
| KL/R-Y=   60.8   |               L4                           FA  =  150.5 |
| KL/R-Z=    8.3   +                                            fa  =    1.8 |
| UNL   = 2250.0   |                   L4  L4                   FCZ =  189.2 |
| C     =  180.0   +                                            FTZ =  224.4 |
| CMY   =   0.45   |                           L4               FCY =  224.4 |
| CMZ   =   0.81   +                               L4  L4       FTY =  224.4 |
| FYLD  =  340.0   |                                     L4     fbz =    0.4 |
| NSF   =    1.0   +---+---+---+---+---+---+---+---+---+---|    fby =    2.2 |
| DFF   =    0.0    0.4                                        FV  =  136.0 |
| dff   =    0.0              ABSOLUTE MZ ENVELOPE             fv  =    0.1 |
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY ( KN-METR)                |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE       0.0         0.0        0.0         0.0         0.0     |
|   LOCATION       0.0         0.0        0.0         0.0         0.0     |
|    LOADING         0           0          0           0           0     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                         DESIGN SUMMARY ( KN-METR)                     *|
|*                         --------------                                *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY              MZ           LOCATION           |
|         ========================================================       |
|          PASS        IS-7.1.1(A)      0.024             4              |
|         10.56 C          -0.2          0.5            2.25             |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
   106      TAP ERED                (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 107 Check 1

```
|-------------------------------------------------------------------------|
|                                                  Y        PROPERTIES    |
|*************                                     |        IN  CM  UNIT  |
|            *  |==============================|   |== ==|     ------------ |
|MEMBER 107  *  |  INDIAN SECTIONS             |   |     |     AX =   18.2  |
|            *  | FR  ISMC75                   |   |     |--Z  AY =    7.2  |
|DESIGN CODE *  |                              |   |     |     AZ =    8.0  |
|  IS-800    *  ================================   |== ==|     SY =   39.1  |
|            *                                                  SZ =   41.6  |
|            *  |<---LENGTH (ME=     1.66 --->|                 RY =    2.9  |
|*************                                                  RZ =    2.9  |
|                                                                         |
|                  0.5( KN-METR)                                          |
|PARAMETER         |L4                                          STRESSES    |
|IN   NEWT  MM     |   L4                                       IN   NEWT  MM|
|---------------   +       L4                              L4   -------------|
| KL/R-Y=   56.5   |                                            FA  =  125.6 |
| KL/R-Z=   56.5   +           L4                       L4      fa  =   15.6 |
| UNL   = 1655.3   |               L4                           FCZ =  156.8 |
| C     =  180.0   +                                 L4          FTZ =  165.0 |
| CMY   =   1.00   |                            L4              FCY =  165.0 |
| CMZ   =   0.40   +                   L4                       FTY =  165.0 |
| FYLD  =  250.0   |                       L4                   fbz =   11.0 |
| NSF   =    1.0   +---+---+---+---+---+---+---+---+---+---|   fby =    8.4 |
| DFF   =    0.0    0.0                                        FV  =  100.0 |
| dff   =    0.0              ABSOLUTE MZ ENVELOPE             fv  =    0.5 |
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY ( KN-METR)                |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE       0.0          0.0        0.0         0.0         0.0    |
|   LOCATION       0.0          0.0        0.0         0.0         0.0    |
|    LOADING         0            0          0           0           0    |
|                                                                         |
|*************************************************************************|
|*                                                                        *|
|*                         DESIGN SUMMARY ( KN-METR)                      *|
|*                         --------------                                 *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY              MZ           LOCATION            |
|         ========================================================        |
|          PASS        IS-7.1.1(A)      0.246             4               |
|         28.36 C           0.3         -0.5           0.00               |
|*                                                                        *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
   107 FR   ISMC75                  (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 108 Check 1

```
|-------------------------------------------------------------------------|
|                                                  Y        PROPERTIES    |
|*************                                     |        IN  CM  UNIT  |
|            *  |==============================|   ===|===     ------------ |
|MEMBER 108  *  |  INDIAN SECTIONS             |      |        AX =   58.4  |
|            *  |     TAP ERED                 |      |   --Z  AY =   35.0  |
|DESIGN CODE *  |                              |      |        AZ =   16.0  |
|  IS-800    *  ================================   ===|===     SY =   80.1  |
|            *                                                  SZ = 1213.4  |
|            *  |<---LENGTH (ME=     0.40 --->|                 RY =    3.7  |
|*************                                                  RZ =   27.0  |
|                                                                         |
|                  0.0( KN-METR)                                          |
|PARAMETER         |L4                                          STRESSES    |
|IN   NEWT  MM     |   L4  L4                                   IN   NEWT  MM|
|---------------   +           L4                               -------------|
| KL/R-Y=   10.8   |               L4                           FA  =  203.4 |
| KL/R-Z=    1.5   +                   L4                      fa  =    0.0 |
| UNL   =  400.0   |                   L4  L4                  FCZ =  224.0 |
| C     =  180.0   +                                           FTZ =  224.4 |
| CMY   =    0.60  |                           L4              FCY =  224.4 |
| CMZ   =    0.60  +                               L4  L4      FTY =  224.4 |
| FYLD  =  340.0   |                                     L0    fbz =    0.0 |
| NSF   =    1.0   +---+---+---+---+---+---+---+---+---+---|   fby =    0.0 |
| DFF   =    0.0   -0.0                                         FV  =  136.0 |
| dff   =    0.0              ABSOLUTE MZ ENVELOPE              fv  =    0.0 |
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY ( KN-METR)                |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE       0.0         0.0        0.0         0.0         0.0     |
|   LOCATION       0.0         0.0        0.0         0.0         0.0     |
|    LOADING         0           0          0           0           0     |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                         DESIGN SUMMARY ( KN-METR)                     *|
|*                         --------------                                *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY              MZ           LOCATION           |
|         ========================================================       |
|           PASS        IS-7.1.1(A)      0.000             1             |
|           0.18 C           0.0          0.0           0.00             |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
   108      TAP ERED                (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 109 Check 1

```
|-------------------------------------------------------------------------|
|                                                    Y        PROPERTIES   |
|*************                                       |        IN  CM  UNIT |
|            *  |==============================|    |== ==|     ------------ |
|MEMBER 109  *  |  INDIAN SECTIONS             |    |     |     AX =   18.2  |
|            *  | FR  ISMC75                   |    |     |--Z  AY =    7.2  |
|DESIGN CODE *  |                              |    |     |     AZ =    8.0  |
|  IS-800    *  ================================    |== ==|     SY =   39.1  |
|            *                                                   SZ =   41.6  |
|            *  |<---LENGTH (ME=     1.66 --->|                  RY =    2.9  |
|*************                                                   RZ =    2.9  |
|                                                                         |
|                  1.6( KN-METR)                                          |
|PARAMETER         |L4                                                    STRESSES     |
|IN   NEWT  MM     |   L4                                                 IN   NEWT  MM|
|---------------   +       L4                                             -------------|
| KL/R-Y=   56.5   |                                                      FA  =  125.6 |
| KL/R-Z=   56.5   +           L4                                         fa  =    4.5 |
| UNL   = 1655.3   |               L4                                    FCZ =  156.8 |
| C     =  400.0   +                                              L4      FTZ =  165.0 |
| CMY   =   0.48   |                    L4              L4               FCY =  165.0 |
| CMZ   =   0.45   +                        L4      L4                    FTY =  165.0 |
| FYLD  =  250.0   |                             L4                      fbz =   38.2 |
| NSF   =    1.0   +---+---+---+---+---+---+---+---+---+---|              fby =    2.2 |
| DFF   =    0.0    0.0                                                   FV  =  100.0 |
| dff   =    0.0              ABSOLUTE MZ ENVELOPE                        fv  =    2.0 |
|                                (WITH LOAD NO.)                                      |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY ( KN-METR)                |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z     |
|                                                                         |
|      VALUE       0.0          0.0        0.0         0.0         0.0     |
|   LOCATION       0.0          0.0        0.0         0.0         0.0     |
|    LOADING         0            0          0           0           0     |
|                                                                         |
|*************************************************************************|
|*                                                                        *|
|*                         DESIGN SUMMARY ( KN-METR)                      *|
|*                         --------------                                 *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY              MZ           LOCATION            |
|         ========================================================        |
|           PASS        IS-7.1.1(A)      0.293             4              |
|           8.19 C          -0.1          1.6            0.00             |
|*                                                                        *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
   109 FR   ISMC75                  (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 110 Check 1

```
|-------------------------------------------------------------------------|
|                                                    Y        PROPERTIES   |
|*************                                       |        IN  CM  UNIT |
|            *  |==============================|    |== ==|     ------------ |
|MEMBER 110  *  |  INDIAN SECTIONS             |    |     |     AX =   18.2  |
|            *  | FR  ISMC75                   |    |     |--Z  AY =    7.2  |
|DESIGN CODE *  |                              |    |     |     AZ =    8.0  |
|  IS-800    *  ================================   |== ==|     SY =   39.1  |
|            *                                                   SZ =   41.6  |
|            *  |<---LENGTH (ME=     2.79 --->|                 RY =    2.9  |
|*************                                                   RZ =    2.9  |
|                                                                         |
|                  0.2( KN-METR)                                          |
|PARAMETER         |                                        L4  STRESSES    |
|IN   NEWT  MM     |L4                                          IN   NEWT  MM|
|----------------  +   L4  L4                                   -------------|
| KL/R-Y=   95.3   |           L4                       L4      FA  =  150.0 |
| KL/R-Z=   95.4   +               L4                         fa  =    7.6 |
| UNL   = 2793.3   |                                 L4       FCZ =  145.8 |
| C     =  180.0   +                                          FTZ =  165.0 |
| CMY   =   0.92   |                   L4                    FCY =  165.0 |
| CMZ   =   0.40   +                           L4             FTY =  165.0 |
| FYLD  =  250.0   |                        L5                fbz =    5.2 |
| NSF   =    1.0   +---+---+---+---+---+---+---+---+---+---|   fby =   14.7 |
| DFF   =    0.0    0.0                                      FV  =  100.0 |
| dff   =    0.0              ABSOLUTE MZ ENVELOPE           fv  =    0.1 |
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY ( KN-METR)                |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE       0.0         0.0        0.0         0.0         0.0     |
|   LOCATION       0.0         0.0        0.0         0.0         0.0     |
|    LOADING         0           0          0           0           0     |
|                                                                         |
|*************************************************************************|
|*                                                                        *|
|*                         DESIGN SUMMARY ( KN-METR)                      *|
|*                         --------------                                 *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY              MZ           LOCATION            |
|         ========================================================        |
|          PASS        IS-7.1.2         0.171             4               |
|         13.88 T           0.6         -0.2            0.00              |
|*                                                                        *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
    110 FR   ISMC75                  (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 111 Check 1

```
|-------------------------------------------------------------------------|
|                                                    Y        PROPERTIES   |
|*************                                       |        IN  CM  UNIT |
|            *  |==============================|    |== ==|     ------------ |
|MEMBER 111  *  |  INDIAN SECTIONS             |    |     |     AX =   18.2  |
|            *  | FR  ISMC75                   |    |     |--Z  AY =    7.2  |
|DESIGN CODE *  |                              |    |     |     AZ =    8.0  |
|  IS-800    *  ================================    |== ==|     SY =   39.1  |
|            *                                                   SZ =   41.6  |
|            *  |<---LENGTH (ME=     2.79 --->|                 RY =    2.9  |
|*************                                                   RZ =    2.9  |
|                                                                         |
|                  0.1( KN-METR)                                          |
|PARAMETER         |                                          L4  STRESSES     |
|IN   NEWT  MM     |                                              IN   NEWT  MM|
|----------------  +L1             L4  L4                         -------------|
| KL/R-Y=   95.3   |                        L4                    FA  =  150.0 |
| KL/R-Z=   95.4   +           L4                                 fa  =   26.0 |
| UNL   = 2793.3   |                                     L4       FCZ =  145.8 |
| C     =  180.0   +   L1  L4                                      FTZ =  165.0 |
| CMY   =   0.62   |                                            FCY =  165.0 |
| CMZ   =   0.67   +                            L4               FTY =  165.0 |
| FYLD  =  250.0   |                                L1          fbz =    1.4 |
| NSF   =    1.0   +---+---+---+---+---+---+---+---+---+---|   fby =   14.4 |
| DFF   =    0.0    0.0                                        FV  =  100.0 |
| dff   =    0.0               ABSOLUTE MZ ENVELOPE            fv  =    0.2 |
|                                 (WITH LOAD NO.)                         |
|                                                                         |
|                      MAX FORCE/ MOMENT SUMMARY ( KN-METR)               |
|                      -------------------------                          |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z     |
|                                                                         |
|      VALUE       0.0          0.0        0.0         0.0         0.0     |
|   LOCATION       0.0          0.0        0.0         0.0         0.0     |
|    LOADING         0            0          0           0           0     |
|                                                                         |
|*************************************************************************|
|*                                                                        *|
|*                         DESIGN SUMMARY ( KN-METR)                      *|
|*                         --------------                                 *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY              MZ           LOCATION            |
|         ========================================================        |
|          PASS        IS-7.1.2         0.269              4              |
|         47.35 T           0.6          0.1             2.79              |
|*                                                                        *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
    111 FR   ISMC75                  (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 112 Check 1

```
|-------------------------------------------------------------------------|
|                                                  Y        PROPERTIES    |
|*************                                     |        IN  CM  UNIT  |
|            *  |==============================|   |== ==|     ------------ |
|MEMBER 112  *  |  INDIAN SECTIONS             |   |     |     AX =   18.2  |
|            *  | FR  ISMC75                   |   |     |--Z  AY =    7.2  |
|DESIGN CODE *  |                              |   |     |     AZ =    8.0  |
|  IS-800    *  ================================   |== ==|     SY =   39.1  |
|            *                                                  SZ =   41.6  |
|            *  |<---LENGTH (ME=     2.79 --->|                 RY =    2.9  |
|*************                                                  RZ =    2.9  |
|                                                                         |
|                  0.1( KN-METR)                                          |
|PARAMETER        |L4                                           STRESSES    |
|IN   NEWT  MM    |                                             IN   NEWT  MM|
|---------------  +   L4                                        -------------|
| KL/R-Y=   95.3  |                                             FA  =  150.0 |
| KL/R-Z=   95.4  +                                        L4   fa  =   41.8 |
| UNL   = 2793.3  |       L4                                    FCZ =  145.8 |
| C     =  180.0  +           L4                       L4       FTZ =  165.0 |
| CMY   =   0.60  |                                L4           FCY =  165.0 |
| CMZ   =   0.87  +               L4          L4                FTY =  165.0 |
| FYLD  =  250.0  |                   L4  L4                    fbz =    3.3 |
| NSF   =    1.0  +---+---+---+---+---+---+---+---+---+---|     fby =   15.1 |
| DFF   =    0.0   0.0                                        FV  =  100.0 |
| dff   =    0.0              ABSOLUTE MZ ENVELOPE            fv  =    0.3 |
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY ( KN-METR)                |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE       0.0          0.0        0.0         0.0         0.0    |
|   LOCATION       0.0          0.0        0.0         0.0         0.0    |
|    LOADING         0            0          0           0           0    |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                         DESIGN SUMMARY ( KN-METR)                     *|
|*                         --------------                                *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY              MZ           LOCATION           |
|         ========================================================       |
|          PASS        IS-7.1.2         0.390              4              |
|         76.03 T          -0.6          0.1             0.00             |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
   112 FR   ISMC75                  (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 113 Check 1

```
|-------------------------------------------------------------------------|
|                                                    Y        PROPERTIES   |
|*************                                       |        IN  CM  UNIT |
|            *  |==============================|    |== ==|     ------------ |
|MEMBER 113  *  |  INDIAN SECTIONS             |    |     |     AX =   18.2  |
|            *  | FR  ISMC75                   |    |     |--Z  AY =    7.2  |
|DESIGN CODE *  |                              |    |     |     AZ =    8.0  |
|  IS-800    *  ================================    |== ==|     SY =   39.1  |
|            *                                                   SZ =   41.6  |
|            *  |<---LENGTH (ME=     2.79 --->|                 RY =    2.9  |
|*************                                                   RZ =    2.9  |
|                                                                         |
|                  1.0( KN-METR)                                          |
|PARAMETER        |L4                                                STRESSES     |
|IN   NEWT  MM    |   L4                                             IN   NEWT  MM|
|---------------  +       L4                                         -------------|
| KL/R-Y=   95.3  |                                                  FA  =  150.0 |
| KL/R-Z=   95.4  +           L4                              L4     fa  =   61.4 |
| UNL   = 2793.3  |               L4                   L4           FCZ =  145.8 |
| C     =  180.0  +                                                  FTZ =  165.0 |
| CMY   =   0.59  |                                 L4               FCY =  165.0 |
| CMZ   =   0.40  +                    L4      L4                    FTY =  165.0 |
| FYLD  =  250.0  |                         L4                       fbz =   16.1 |
| NSF   =    1.0  +---+---+---+---+---+---+---+---+---+---|          fby =   27.3 |
| DFF   =    0.0   0.1                                             FV  =  100.0 |
| dff   =    0.0              ABSOLUTE MZ ENVELOPE                 fv  =    1.0 |
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY ( KN-METR)                |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z     |
|                                                                         |
|      VALUE       0.0         0.0        0.0         0.0         0.0     |
|   LOCATION       0.0         0.0        0.0         0.0         0.0     |
|    LOADING         0           0          0           0           0     |
|                                                                         |
|*************************************************************************|
|*                                                                        *|
|*                         DESIGN SUMMARY ( KN-METR)                      *|
|*                         --------------                                 *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY              MZ           LOCATION            |
|        ========================================================          |
|          PASS        IS-7.1.2         0.673              4               |
|        111.73 T          -1.1          0.7             2.79              |
|*                                                                        *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
   113 FR   ISMC75                  (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 114 Check 1

```
|-------------------------------------------------------------------------|
|                                                    Y        PROPERTIES   |
|*************                                       |        IN  CM  UNIT |
|            *  |==============================|    |== ==|     ------------ |
|MEMBER 114  *  |  INDIAN SECTIONS             |    |     |     AX =   18.2  |
|            *  | FR  ISMC75                   |    |     |--Z  AY =    7.2  |
|DESIGN CODE *  |                              |    |     |     AZ =    8.0  |
|  IS-800    *  ================================    |== ==|     SY =   39.1  |
|            *                                                   SZ =   41.6  |
|            *  |<---LENGTH (ME=     2.79 --->|                 RY =    2.9  |
|*************                                                   RZ =    2.9  |
|                                                                         |
|                  1.0( KN-METR)                                          |
|PARAMETER        |L4                                                     |
|IN   NEWT  MM    |   L4                                                  |
|---------------  +       L4                                              |
| KL/R-Y=   95.3  |                                                       |
| KL/R-Z=   95.4  +           L4                              L4          |
| UNL   = 2793.3  |               L4                   L4                 |
| C     =  180.0  +                                                       |
| CMY   =   0.59  |                                 L4                    |
| CMZ   =   0.40  +                    L4       L4                        |
| FYLD  =  250.0  |                         L4                            |
| NSF   =    1.0  +---+---+---+---+---+---+---+---+---+---|               |
| DFF   =    0.0   0.1                                                    |
| dff   =    0.0              ABSOLUTE MZ ENVELOPE                        |
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY ( KN-METR)                |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE       0.0         0.0        0.0         0.0         0.0    |
|   LOCATION       0.0         0.0        0.0         0.0         0.0    |
|    LOADING         0           0          0           0           0     |
|                                                                         |
|*************************************************************************|
|*                                                                        *|
|*                         DESIGN SUMMARY ( KN-METR)                      *|
|*                         --------------                                 *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY              MZ           LOCATION            |
|        ==========================================================      |
|          PASS        IS-7.1.2         0.673              4              |
|        111.73 T           1.1          0.7             2.79              |
|*                                                                        *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
    114 FR   ISMC75                  (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 115 Check 1

```
|-------------------------------------------------------------------------|
|                                                  Y        PROPERTIES    |
|*************                                     |        IN  CM  UNIT  |
|            *  |==============================|   |== ==|     ------------ |
|MEMBER 115  *  |  INDIAN SECTIONS             |   |     |     AX =   18.2  |
|            *  | FR  ISMC75                   |   |     |--Z  AY =    7.2  |
|DESIGN CODE *  |                              |   |     |     AZ =    8.0  |
|  IS-800    *  ================================   |== ==|     SY =   39.1  |
|            *                                                  SZ =   41.6  |
|            *  |<---LENGTH (ME=     2.79 --->|                 RY =    2.9  |
|*************                                                  RZ =    2.9  |
|                                                                         |
|                  0.1( KN-METR)                                          |
|PARAMETER        |L4                                           STRESSES    |
|IN   NEWT  MM    |                                             IN   NEWT  MM|
|---------------  +   L4                                        -------------|
| KL/R-Y=   95.3  |                                             FA  =  150.0 |
| KL/R-Z=   95.4  +                                        L4   fa  =   41.8 |
| UNL   = 2793.3  |       L4                                    FCZ =  145.8 |
| C     =  180.0  +           L4                       L4       FTZ =  165.0 |
| CMY   =   0.60  |                                L4           FCY =  165.0 |
| CMZ   =   0.87  +               L4          L4                FTY =  165.0 |
| FYLD  =  250.0  |                   L4  L4                    fbz =    3.3 |
| NSF   =    1.0  +---+---+---+---+---+---+---+---+---+---|     fby =   15.1 |
| DFF   =    0.0   0.0                                        FV  =  100.0 |
| dff   =    0.0              ABSOLUTE MZ ENVELOPE            fv  =    0.3 |
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY ( KN-METR)                |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE       0.0          0.0        0.0         0.0         0.0    |
|   LOCATION       0.0          0.0        0.0         0.0         0.0    |
|    LOADING         0            0          0           0           0    |
|                                                                         |
|*************************************************************************|
|*                                                                       *|
|*                         DESIGN SUMMARY ( KN-METR)                     *|
|*                         --------------                                *|
|*                                                                       *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/         *|
|         FX              MY              MZ           LOCATION           |
|         ========================================================       |
|          PASS        IS-7.1.2         0.390             4              |
|         76.03 T           0.6          0.1            0.00              |
|*                                                                       *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
    115 FR   ISMC75                  (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 116 Check 1

```
|-------------------------------------------------------------------------|
|                                                    Y        PROPERTIES   |
|*************                                       |        IN  CM  UNIT |
|            *  |==============================|    |== ==|     ------------ |
|MEMBER 116  *  |  INDIAN SECTIONS             |    |     |     AX =   18.2  |
|            *  | FR  ISMC75                   |    |     |--Z  AY =    7.2  |
|DESIGN CODE *  |                              |    |     |     AZ =    8.0  |
|  IS-800    *  ================================    |== ==|     SY =   39.1  |
|            *                                                   SZ =   41.6  |
|            *  |<---LENGTH (ME=     2.79 --->|                 RY =    2.9  |
|*************                                                   RZ =    2.9  |
|                                                                         |
|                  0.1( KN-METR)                                          |
|PARAMETER         |                                          L4  STRESSES     |
|IN   NEWT  MM     |                                              IN   NEWT  MM|
|----------------  +L1             L4  L4                         -------------|
| KL/R-Y=   95.3   |                        L4                    FA  =  150.0 |
| KL/R-Z=   95.4   +           L4                                 fa  =   26.0 |
| UNL   = 2793.3   |                                     L4       FCZ =  145.8 |
| C     =  180.0   +   L1  L4                                      FTZ =  165.0 |
| CMY   =   0.62   |                                            FCY =  165.0 |
| CMZ   =   0.67   +                            L4               FTY =  165.0 |
| FYLD  =  250.0   |                                L1          fbz =    1.4 |
| NSF   =    1.0   +---+---+---+---+---+---+---+---+---+---|   fby =   14.4 |
| DFF   =    0.0    0.0                                        FV  =  100.0 |
| dff   =    0.0               ABSOLUTE MZ ENVELOPE            fv  =    0.2 |
|                                 (WITH LOAD NO.)                         |
|                                                                         |
|                      MAX FORCE/ MOMENT SUMMARY ( KN-METR)               |
|                      -------------------------                          |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z     |
|                                                                         |
|      VALUE       0.0          0.0        0.0         0.0         0.0     |
|   LOCATION       0.0          0.0        0.0         0.0         0.0     |
|    LOADING         0            0          0           0           0     |
|                                                                         |
|*************************************************************************|
|*                                                                        *|
|*                         DESIGN SUMMARY ( KN-METR)                      *|
|*                         --------------                                 *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY              MZ           LOCATION            |
|         ========================================================        |
|          PASS        IS-7.1.2         0.269              4              |
|         47.35 T          -0.6          0.1             2.79              |
|*                                                                        *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
   116 FR   ISMC75                  (INDIAN SECTIONS)
```

Steel Design (Track 2) Beam 117 Check 1

```
|-------------------------------------------------------------------------|
|                                                    Y        PROPERTIES   |
|*************                                       |        IN  CM  UNIT |
|            *  |==============================|    |== ==|     ------------ |
|MEMBER 117  *  |  INDIAN SECTIONS             |    |     |     AX =   18.2  |
|            *  | FR  ISMC75                   |    |     |--Z  AY =    7.2  |
|DESIGN CODE *  |                              |    |     |     AZ =    8.0  |
|  IS-800    *  ================================   |== ==|     SY =   39.1  |
|            *                                                   SZ =   41.6  |
|            *  |<---LENGTH (ME=     2.79 --->|                 RY =    2.9  |
|*************                                                   RZ =    2.9  |
|                                                                         |
|                  0.2( KN-METR)                                          |
|PARAMETER         |                                        L4  STRESSES    |
|IN   NEWT  MM     |L4                                          IN   NEWT  MM|
|----------------  +   L4  L4                                   -------------|
| KL/R-Y=   95.3   |           L4                       L4      FA  =  150.0 |
| KL/R-Z=   95.4   +               L4                         fa  =    7.6 |
| UNL   = 2793.3   |                                 L4       FCZ =  145.8 |
| C     =  180.0   +                                          FTZ =  165.0 |
| CMY   =   0.92   |                   L4                    FCY =  165.0 |
| CMZ   =   0.40   +                           L4             FTY =  165.0 |
| FYLD  =  250.0   |                        L5                fbz =    5.2 |
| NSF   =    1.0   +---+---+---+---+---+---+---+---+---+---|   fby =   14.7 |
| DFF   =    0.0    0.0                                      FV  =  100.0 |
| dff   =    0.0              ABSOLUTE MZ ENVELOPE           fv  =    0.1 |
|                                (WITH LOAD NO.)                          |
|                                                                         |
|                     MAX FORCE/ MOMENT SUMMARY ( KN-METR)                |
|                     -------------------------                           |
|                                                                         |
|                AXIAL      SHEAR-Y    SHEAR-Z    MOMENT-Y    MOMENT-Z   |
|                                                                         |
|      VALUE       0.0         0.0        0.0         0.0         0.0     |
|   LOCATION       0.0         0.0        0.0         0.0         0.0     |
|    LOADING         0           0          0           0           0     |
|                                                                         |
|*************************************************************************|
|*                                                                        *|
|*                         DESIGN SUMMARY ( KN-METR)                      *|
|*                         --------------                                 *|
|*                                                                        *|
|*       RESULT/     CRITICAL COND/     RATIO/         LOADING/          *|
|         FX              MY              MZ           LOCATION            |
|         ========================================================        |
|          PASS        IS-7.1.2         0.171             4               |
|         13.88 T          -0.6         -0.2           0.00               |
|*                                                                        *|
|*************************************************************************|
|                                                                         |
|-------------------------------------------------------------------------|
   117 FR   ISMC75                  (INDIAN SECTIONS)
```

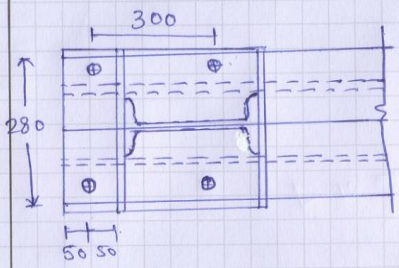
5.2 Steel connections

Design of beam column connection and hanger beam connection are done as given below

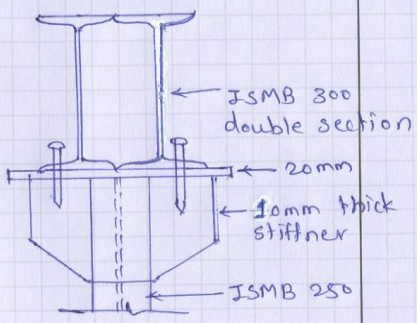
Project Name Megadoor Supporting frame Sheet No. 1/1

Part of Structure Beam - Column connection Date

Design Checked by Job Ref



Plan



Elevation

The maximum design moment in beam is 32 KN·m

$$\therefore \text{tension / bolt} = \left\{\frac{32}{L \cdot A}\right\} / 2$$

$$= \frac{32}{0.3} / 2$$

$$= 53.3 \text{ KN}$$

Bolt value of M20 bolt in tension = 88.2 KN ---- (Ref. IS 4000 Grade 8.8)

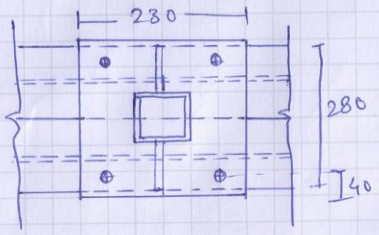
∴ 4 bolts of 20 mm diameter is to be provided.

E-mail : enquiry@designwerkz.in Web : www.designwerkz.in

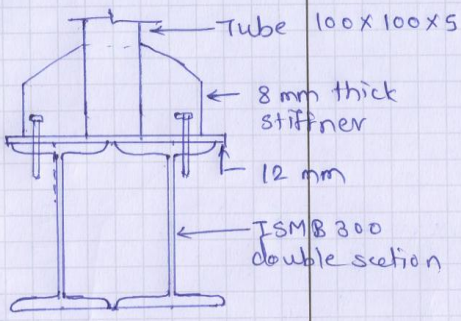
Project Name Megadoor Supporting frame Sheet No. 1/1

Part of Structure Hanger Connection Date

Design Checked by Job Ref



Plan



Elevation

Maximum tension occured in hanger = 21 kN

$\therefore$ tension / bolt $= \frac{21}{4} = 5.25$ kN

Bolt value of M16 bolt in tension = 56.7 kN -- [Ref IS 4000 Grade 8.8]

$\therefore$ 4 bolts of 16 mm diameter are to be provided.